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22 May 1963
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30 October 1961

MILITARY STANDARD

GRAPHIC SYMBOLS FOR ELECTRICAL AND ELECTRONICS DIAGRAMS

UNITED STATES
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Graphic Symbols for Electrical and Electronics Diagrams

MIL-STD-15-1A

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, and the Defense Supply Agency, effective 22 May 1963.

2. Recommended corrections, additions or deletions should be addressed to the Chief, Bureau of Ships, Department of the Navy, Washington 25, D. C. Attention: Specifications and Standards Branch, Code 621, with the following information:

(a) Number and title of this standard.

(b) Item for which changed, corrected or new symbol is requested; type or class of item, frequency of use (per drawing and per equipment).

(c) Present symbol used and authority for this use (if governmental, specify agency; if commercial, specify company).

(d) Proposed symbol.

(e) Reasons for recommending change (attach source material if feasible).

(f) Reference designation which is used with symbol.

(g) Any other known industry or agency usage of same or similar symbols.

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FOREWORD

Standard MIL-STD-15-1A contains a listing of graphic drawing symbols for electrical and electronics parts and recommendations as to their use. These symbols should be used by all who have occasion to prepare electrical or electronics schematic drawings for the Departments of the Army, the Navy, and the Air Force, and the Defense Supply Agency.

During the preparation of Standard MIL-STD-15-1A it was recognized that a need existed for the coordination of drafting symbols and drafting practices used by the Military, to make them compatible with those in private industry and accepted nationwide industry standards.

The symbols contained in Standard MIL-STD-15-1 were considered carefully in the preparation of Standard MIL-STD-15-1A. In many instances symbols generally accepted in practice were selected even though more definitive pictorial symbols could have been devised. Certain new symbols have been included to cover items which have come into use since Standard MIL-STD-15-1 was adopted, or to more clearly identify given types of items.

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GRAPHIC SYMBOLS FOR ELECTRICAL AND ELECTRONICS DIAGRAMS

1. SCOPE

1.1 ***Purpose and scope.*** This Standard provides a list of graphic symbols for use on electrical and electronics schematic diagrams.

1.2 ***Use of symbols.*** Graphic symbols for electrical engineering are a shorthand used to show graphically the functioning or interconnections of a circuit. *A graphic symbol represents the function of a part in the circuit.* Graphic symbols are used on single-line (one-line) diagrams, on schematic or elementary diagrams or as applicable on connection or wiring diagrams. Graphic symbols are correlated with parts lists, descriptions, or instructions by means of reference designations.

1.3 ***Arrangement.*** All terms appear in the index. In the index, "item" refers to the number assigned a symbol in the list of symbols. Items are arranged alphabetically in order of family-group names. Terms in preferred usage and current alternates are listed. The symbol F indicates standard item names from the Federal Item Identification Guide, Handbook H6-1, and are included wherever applicable.

1.4 In the list, graphic symbols appear directly under their respective family names. Single-line (one-line) symbols appear at the left in each column. Complete symbols appear at the right of the column. Symbols suitable for both purposes are centered in each column.

1.5 Symbols that have been recommended by the International Electrotechnical Commission are indicated by IEC.

1.6 Where alternative symbols are shown, the relative position of the symbols does not necessarily imply a preference.

1.7 ***Application.*** An application is an example of a combination of symbols in the list. No attempt has been made to list all possible applications. Additional applications may be devised provided they are a reasonable and intelligible use of the symbols in the list.

1.7.1 For application of these symbols in electrical and electronics diagrams see American Drafting Standard Manual ASA-Y14.15.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-12—Abbreviations for Use on Drawings and in Technical-Type Publications.

MIL-STD-15-2—Electrical Wiring Equipment Symbols for Ships' Plans.

MIL-STD-15-3—Electrical Wiring Symbols for Architectural and Electrical Layout Drawings.

MIL-STD-16—Electrical and Electronic Reference Designations.

MIL-STD-27—Designations for Electric Switchgear and Control Devices.

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2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION

C37.2—Automatic Station Control, Supervisory, and Associated Telemetering Equipments.

Y14.15—American Drafting Standard Manual, Electrical and Electronics Diagrams.

(Application for copies should be addressed to the American Standards Association Incorporated, 10 East 40th Street, New York 16, N. Y.)

3. DEFINITIONS AND APPLICABLE DRAFTING PRACTICES

3.1 Single-line (one-line diagrams). A diagram which shows, by means of single lines and graphic symbols, the course of an electric circuit or system of circuits and the component devices or parts used therein.

3.2 Schematic or elementary diagrams. A diagram which shows, by means of graphic symbols, the electrical connections and functions of a specific circuit arrangement. The schematic diagram facilitates tracing the circuit and its functions without regard to the actual physical size, shape, or location of the component device or parts.

3.3 A symbol shall be considered as the aggregate of all its parts.

3.4 The orientation of a symbol on a drawing, including a mirror image presentation, does not alter the meaning of the symbol. Letters and numbers that constitute a part of a symbol shall not be presented in mirror image form.

3.5 The width of a line does not affect the meaning of the symbol. In specific cases, a wider line may be used for emphasis.

3.6 The symbols shown in this standard are in their correct relative size. This relationship shall be maintained as nearly as possible on any particular drawing, regardless of the size of the symbol used.

3.7 A symbol may be drawn to any proportional size that suits a particular drawing, depending on reduction or enlargement anticipated. If essential for purposes of contrast, some symbols may be drawn relatively smaller than the other symbols on a diagram. It is recommended that only two sizes be used on any one diagram.

3.8 The arrowhead of a symbol may be closed → or open ⟶ unless otherwise noted in this standard.

3.9 The standard symbol for a TERMINAL (∘) may be added to each point of attachment to the connecting lines to any one of the graphic symbols. Such added terminal symbols should not be considered as part of the individual graphic symbol, unless the terminal symbol is included in the symbol shown in this standard.

3.10 For simplification of a diagram, parts of a symbol for a device, such as a relay or contactor, may be separated. If this is done, provide suitable designations to show proper correlation of the parts.

3.11 In general, the angle at which a connecting line is brought to a graphic symbol has no particular significance unless otherwise noted in this standard.

3.12 Associated or future paths and equipment shall be shown by lines composed of short dashes: - - - .

3.13 Details of type, impedance, rating and so forth may be added when required, adjacent to any symbol. If used, abbreviations shall be in accordance with Standard MIL-STD-12. Letter combinations used as parts of graphic symbols are not abbreviations.

3.14 Reference designations shall be assigned and used in accordance with Standard MIL-STD-16.

3.15 Symbols for ships' plans shall be in accordance with Standard MIL-STD-15-2.

3.16 Symbols for architectural and layout drawings shall be in accordance with Standard MIL-STD-15-3, supplemented by symbols herein for items not contained in Part 3.

4. GENERAL REQUIREMENTS

4.1 Not applicable.

5. LIST OF SYMBOLS AND DETAILS

5.1 The graphic symbols (items) in this list are arranged by item number indicating a generic class. The item number is not significant other than to distinguish between classes. The initial decimal portion of the complete item number indicates a further breakdown of symbols by type or functional designation, usually in alphabetical sequence. Further decimal subordinations indicate a symbol designed for a specific single function arranged alphabetically in order of the name modifiers. Thus the use of this decimal, item-numbering system for each symbol permits changes and revisions, or insertions, to be made conveniently, and without regard for paragraph and section numbering.

5.2 To locate the symbol for a specific part, find the item number under the colloquial or functional name in the index.

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LIST OF SYMBOLS

1 ADJUSTABILITY VARIABILITY

These recognition symbols shall be drawn at about 45° across the body of the symbol to which they are applied. For typical applications see 2.2.3, 5, 8.1.3, 68.8.

Note 1A—Use only if essential to indicate special property.

1.1 Adjustability (extrinsic adjustability).

1.1.1 General


IEC

1.1.2 Preset, general


IEC See note 1A

1.1.3 Linear (shown applied to 1.1.1)


IEC See note 1A

1.1.4 Nonlinear (shown applied to 1.1.1)


IEC See note 1A

1.2 Inherent variability (intrinsic variability)

1.2.1 Linear


IEC See note 1A

1.2.2 Nonlinear


IEC See note 1A

1.3 Special features (shown applied to the general adjustability symbol)

1.3.1 Continuous


IEC See note 1A

1.3.2 In steps


IEC See note 1A

1.4 Special features (shown applied to the general preset symbol)

1.4.1 Continuous


IEC See note 1A

1.4.2 In steps


IEC See note 1A

2 AMPLIFIER E

See also MACHINE, ROTATING (items 46.8.20 to 46.8.23).

2.1 General

The triangle is pointed in the direction of transmission.

The symbol represents any method of amplification (electron tube, solid state device, magnetic device, etc.).

Note 2.1A—If identification, electrical values, location data and similar information must be noted within a symbol the size or aspect ratio of the original symbol may be altered providing its distinctive shape is retained.

Amplifier use may be indicated in the triangle by words, standard abbreviations, or a letter combination from the following list.

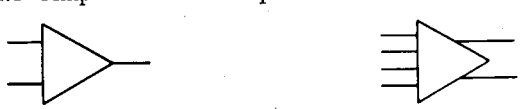
BDG	Bridging	MON	Monitoring
BST	Booster	PGM	Program
CMP	Compression	PRE	Preliminary
DC	Direct Current	PWR	Power
EXP	Expansion	TRQ	Torque
LIM	Limiting		



See note 2.1A

2.2 Applications

2.2.1 Amplifier with two inputs



See note 2.1A

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

2.2.2 Amplifier with two outputs



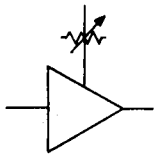
See note 2.1A

2.2.3 Amplifier with adjustable gain



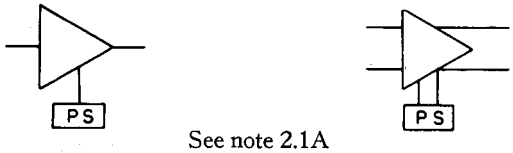
See note 2.1A

2.2.4 Amplifier with associated attenuator



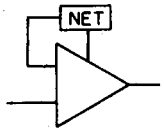
See note 2.1A

2.2.5 Amplifier with associated power supply



See note 2.1A

2.2.6 Amplifier with external feedback path



See note 2.1A

3 ANTENNA F

3.1 General

Types or functions may be indicated by words or abbreviations adjacent to the symbol for clarity.



3.1.1 Dipole



3.1.2 Loop



3.2 Counterpoise, antenna F



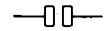
4 ARRESTER, LIGHTNING F
ARRESTER (Electric surge, etc.)
GAP

4.1 General

4.2 Carbon block

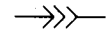
Block, telephone protector F

The sides of the rectangle are to be approximately in the ratio of 1 to 2 and the space between rectangles shall be approximately equal to the width of a rectangle.

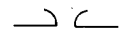


4.3 Electrolytic or aluminum cell

This symbol is not composed of arrowheads.

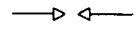


4.4 Horn gap

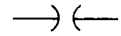


4.5 Protective gap

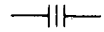
These triangles shall not be filled.



4.6 Sphere gap



4.7 Valve or film element



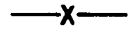
4.8 Multigap, general

4.9 Application: gap plus valve plus ground, 2 pole



4.10 Intentional isolation of DC path in coaxial or waveguide applications

Commonly used for coaxial and waveguide transmission.



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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5 ATTENUATOR (ATTENUATOR, VARIABLE F)

See also PAD (item 57).

5.1 General



5.2 Balanced, general





5.3 Unbalanced, general

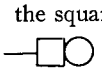




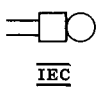
6 AUDIBLE SIGNALING DEVICE

6.1 Bell, electrical F; ringer, telephone F

Note 6.1A—If specific identification is required, the abbreviation AC or DC may be added within the square.



See note 6.1A



6.2 Buzzer F



See note 6.1A



6.3 Horn, electrical F; loudspeaker F; siren F; underwater sound projector or transceiver

6.3.1 General

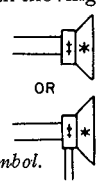




6.3.2 If specific identification of loudspeaker parts is required, the following letter combinations may be added in the symbol at the locations indicated by the * and the ‡

*HN	Horn, electrical F
*HW	Howler
*LS	Loudspeaker F
*SN	Siren F
‡EM	Electromagnetic with moving coil (moving coil leads should be identified)
‡EMN	Electromagnetic with moving coil and neutralizing winding (moving coil leads should be identified)
‡MG	Magnetic armature
‡PM	Permanent magnet with moving coil





*‡ *The * and ‡ are not part of the symbol.*

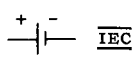
6.4 Sounder, telegraph F



7 BATTERY F

The long line is always positive, but polarity may be indicated in addition.

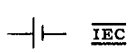
Example:



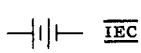
7.1 Generalized direct-current source



7.2 One cell



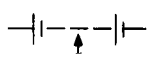
7.3 Multicell



7.3.1 Multicell battery with 3 taps



7.3.2 Multicell battery with adjustable tap



Note—Single-line (one-line) symbols appear at the left, comple symbols at the right, and symbols suitable for both purposes are centered in each column.

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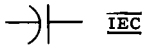
LIST OF SYMBOLS

8 CAPACITOR E

See also TERMINATION (item 82.4).

8.1 General

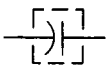
If it is necessary to identify the capacitor electrodes, the curved element shall represent the outside electrode in fixed paper-dielectric and ceramic-dielectric capacitors, the moving element in adjustable and variable capacitors, and the low-potential element in feed-through capacitors.



8.1.1 Polarized capacitor

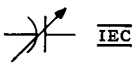


8.1.2 Application: shielded capacitor

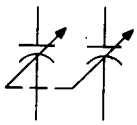


8.1.3 Application: adjustable or variable capacitor

If it is necessary to identify trimmer capacitors, the letter T should appear adjacent to the symbol.

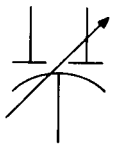


8.1.4 Application: adjustable or variable capacitors with mechanical linkage of units

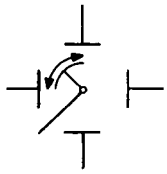


8.2 Continuously adjustable or variable differential capacitor

The capacitance of one part increases as the capacitance of the other part decreases.



8.2.1 Phase-shifter capacitor



8.3. Split-stator capacitor

The capacitances of both parts increase simultaneously.



8.4 Shunt capacitor



8.4.1 Coupling capacitor (for power line carrier)

Note 8.4.1A—The asterisk is not part of the symbol. If specific identification is desired, the asterisk is to be replaced by one of the following letter combinations.

COM	Carrier communication
LC	Carrier load control
REL	Carrier relaying
SUP	Carrier supervisory
TLM	Carrier telemetering
TT	Carrier transferred trip



* See note 8.4.1A

8.5 Feed-through capacitor (with terminals shown on feed-through element for clarity)

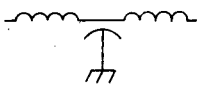
Commonly used for bypassing high-frequency currents to chassis.



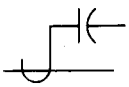
Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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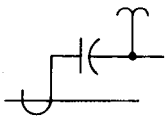
8.5.1 Application: feed-through capacitor between 2 inductors with third lead connected to chassis



8.6 Capacitance bushing for circuit breaker or transformer

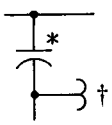


8.6.1 Application: capacitance-bushing potential device



8.7 Application: Carrier coupling capacitor potential device (used to provide a power-system-frequency voltage and also coupling for carrier signals)

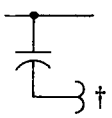
Note 8.7A—The dagger is not a part of the symbol. If specific indication is desired, the dagger is to be replaced by a letter combination from item 48, note 48A.



* See note 8.4.1A

† See note 8.7A

8.7.1 Application: Coupling capacitor potential device (used only to provide a power-system frequency voltage)



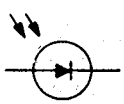
† See note 8.7A

9 CELL, PHOTOSENSITIVE (Semiconductor)

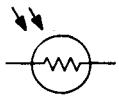
See also PHOTOTUBE (item 34.11.6).

9.1 Asymmetrical photoconductive transducer

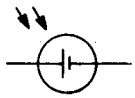
The equilateral triangle within the circle shall be filled.



9.2 Symmetrical photoconductive transducer (resistive)



9.3 Photovoltaic transducer; barrier photocell; blocking-layer cell; solar cell



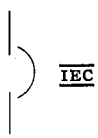
10 CHASSIS CONNECTION
FRAME CONNECTION

See CIRCUIT RETURN (item 13.2).

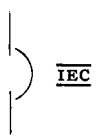
11 CIRCUIT BREAKER F

If it is desired to show the condition causing the breaker to trip, the relay-protective-function symbols in item 66.6 may be used alongside the breaker symbol.

11.1 General



11.2 Air circuit breaker, if distinction is needed; for alternating-current circuit breakers rated at 1,500 volts or less and for all direct-current circuit breakers

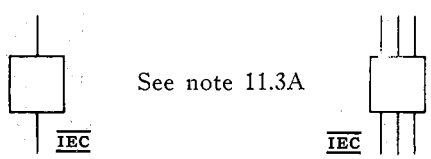


11.2.1 Network protector



11.3 Circuit breaker, other than covered by item 11.2 The symbol in the right column is for a 3-pole breaker.

Note 11.3A—On a power diagram, the symbol may be used without other identification. On a composite drawing where confusion with the general circuit element symbol (item 12) may result, add the identifying letters CB inside or adjacent to the square.

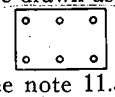


Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

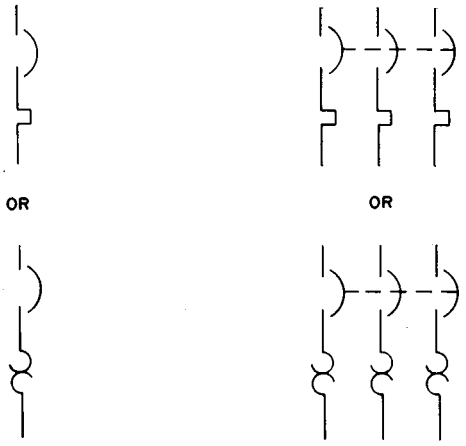
11.3.1 On a connection or wiring diagram, a 3-pole single-throw circuit breaker (with terminals shown) may be drawn as shown below



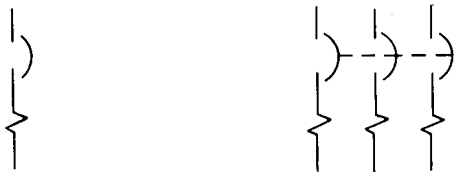
See note 11.3A
FOR CONNECTION OR WIRING DIAGRAM

11.4 Applications

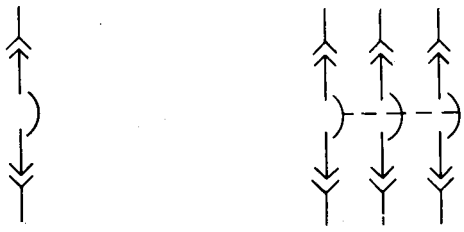
11.4.1 3-pole circuit breaker with thermal overload device in all 3 poles



11.4.2 3-pole circuit breaker with magnetic overload device in all 3 poles



11.4.3 3-pole circuit breaker, drawout type



12 **CIRCUIT ELEMENT** (General)

Note 12A—The asterisk is not a part of the symbol. Always indicate the type of apparatus by appropriate words or letters in the rectangle.

Note 12B—If identification, electrical values, location data and similar information must be noted within a symbol the size or the proportion of the original symbol may be altered providing its distinctive shape is retained.



*See note 12A

12.1 Accepted abbreviations in the latest edition of Standard MIL-STD-12 may be used in the rectangle.

12.2 The following letter combinations may be used in the rectangle.

EQ	Equalizer
FAX	Facsimile set F
FL	Filter
FL-BE	Filter, band elimination
FL-BP	Filter, band pass F
FL-HP	Filter, high pass F
FL-LP	Filter, low pass F
PS	Power supply F
RG	Recording unit
RU	Reproducing unit
DIAL	Telephone dial
TEL	Telephone station
TPR	Teleprinter F
TTY	Teletypewriter F

12.3 Additional letter combinations as follows may be employed, but the use of specific graphic symbols included elsewhere in this standard is preferred.

AR	Amplifier F
AT	Attenuator
C	Capacitor F
CB	Circuit breaker F
HS	Handset F
I	Indicating or switchboard lamp
L	Inductor
J	Jack
LS	Loudspeaker F
MIC	Microphone F
OSC	Oscillator
PAD	Pad
P	Plug
HT	Receiver, headset
K	Relay F
R	Resistor F
S	Switch F or key switch
T	Transformer F
WR	Wall receptacle

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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13 CIRCUIT RETURN

13.1 Ground

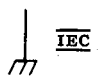
(*A*) A direct conducting connection to the earth or body of water that is a part thereof.

(*B*) A conducting connection to a structure that serves a function similar to that of an earth ground (that is, a structure such as a frame of an air, space, or land vehicle that is not conductively connected to earth).



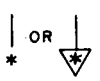
13.2 Chassis or frame connection

A conducting connection to a chassis or frame of a unit. The chassis or frame may be at a substantial potential with respect to the earth or structure in which this chassis or frame is mounted.



13.3 Common connections

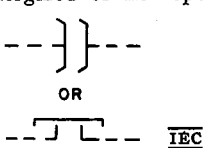
Conducting connections made to one another. All like-designated points are connected.



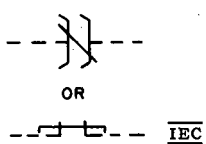
* The asterisk is not a part of the symbol. Identifying values, letters, numbers, or marks shall replace the asterisk. For the triangular symbol this identification shall be placed within the triangle or, if essential for legibility, adjacent to the triangle.

14 CLUTCH
BRAKE

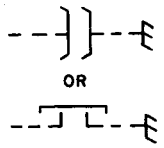
14.1 Clutch disengaged when operating means (not shown) is de-energized or non-operated



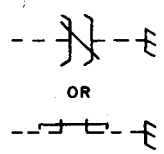
14.2 Clutch engaged when operating means (not shown) is de-energized or non-operated



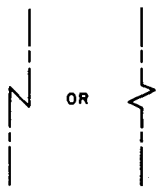
14.3 Brake applied when operating means (not shown) is energized



14.4 Brake released when operating means (not shown) is energized



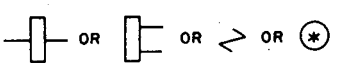
15 ‡COIL, MAGNETIC BLOWOUT



16 COIL, OPERATING
COIL, RELAY

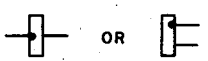
See also item 42; INDUCTOR; WINDING (Machine or transformer); REACTOR.

Note 16A—The asterisk is not a part of the symbol. Always replace the asterisk by a device designation. (See for example American Standard C37.2 or MIL-STD-27.)



* See note 16A

16.1 Semicircular dot indicates inner end of winding



16.2 Application: multiwinding coil (2 windings shown)

Note 16.2A—The ends of a given winding shall be shown directly opposite each other on opposite sides of the core, or adjacent to each other on the same side of the core.



See note 16.2A

‡*The broken line - —— indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

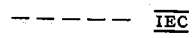
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LIST OF SYMBOLS

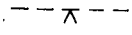
17 CONNECTION, MECHANICAL INTERLOCK, MECHANICAL

The preferred location of the mechanical connection is as shown in the various applications, but other locations may be equally acceptable.

17.1 Mechanical connection (*short dashes*)



17.2 Mechanical connection or interlock with fulcrum (*short dashes*)



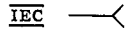
17.3 Mechanical interlock, other

INDICATE BY A NOTE

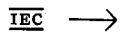
18 CONNECTOR DISCONNECTING DEVICE

The connector symbol is not an arrowhead. It is larger and the lines are drawn at a 90-degree angle.

18.1 Female contact

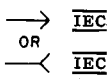


18.2 Male contact



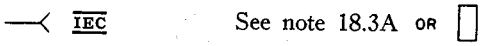
18.3 Connector assembly, movable or stationary portion; jack, plug, or receptacle

Note 18.3A—Use appropriate number of contact symbols. IEC

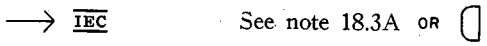


See note 18.3A

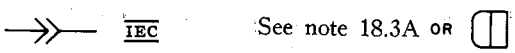
18.3.1 Commonly used for a jack or receptacle (usually stationary)



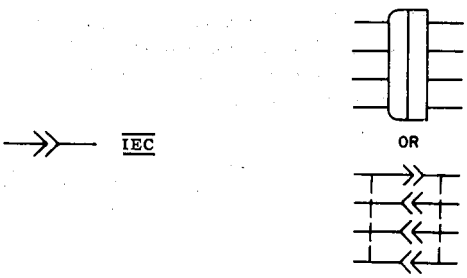
18.3.2 Commonly used for a plug (usually movable)



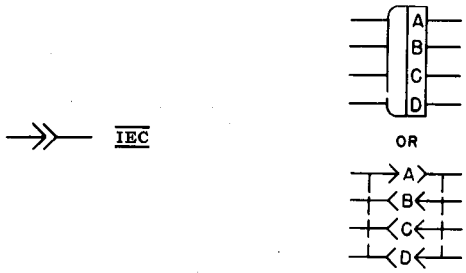
18.4 Separable connectors (engaged)



18.4.1 Application: engaged 4-conductor connectors; the plug has 1 male and 3 female contacts



18.4.2 Application: engaged 4-conductor connectors, the plug has 1 male and 3 female contacts with individual contact designations shown in the complete-symbol column



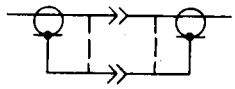
18.5 Coaxial connectors

18.5.1 Engaged coaxial and waveguide connectors

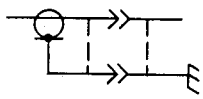
Coaxial recognition sign may be added if necessary. See PATH, TRANSMISSION (item 58.8.2)



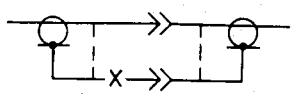
18.5.1.1 Coaxial with the outside conductor shown carried through



18.5.1.2 Coaxial with outside conductor shown carried through; with outside conductor terminated on chassis



18.5.1.3 Coaxial with center conductor shown carried through; outside conductor not carried through



18.6 Communication switchboard-type connector

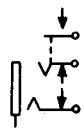
18.6.1 2-conductor (jack)



18.6.2 2-conductor (plug)



18.6.3 3-conductor (jack) with 2 break contacts (normals) and 1 auxiliary make contact



18.6.4 3-conductor (plug)



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

18.7 Communication switchboard-type connector with circuit normalled through

"Normalled" indicates that a through circuit may be interrupted by an inserted connector. As shown here, the inserted connector opens the through circuit and connects to the circuit towards the left.

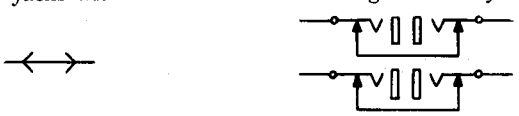
Items 18.7.1 through 18.7.4 show 2-conductor jacks. The "normal" symbol is applicable to other types of connectors.



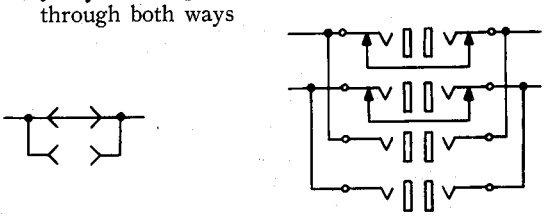
18.7.1 Jacks with circuit normalled through one way



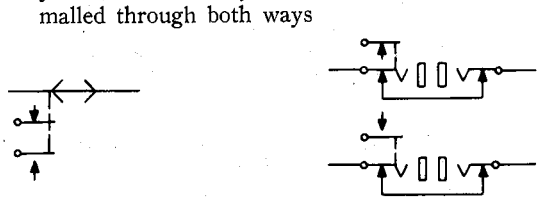
18.7.2 Jacks with circuit normalled through both ways



18.7.3 Jacks in multiple, one set with circuit normalled through both ways



18.7.4 Jacks with auxiliary contacts, with circuit normalled through both ways



18.8 Connectors of the type commonly used for power-supply purposes (convenience outlets and mating connectors)

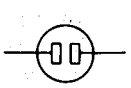
18.8.1 Female contact



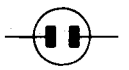
18.8.2 Male contact



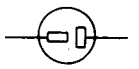
18.8.3 2-conductor nonpolarized connector with female contacts



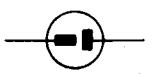
18.8.4 2-conductor nonpolarized connector with male contacts



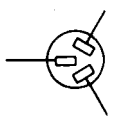
18.8.5 2-conductor polarized connector with female contacts



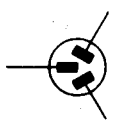
18.8.6 2-conductor polarized connector with male contacts



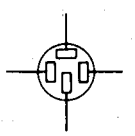
18.8.7 3-conductor polarized connector with female contacts



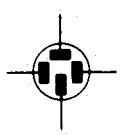
18.8.8 3-conductor polarized connector with male contacts



18.8.9 4-conductor polarized connector with female contacts



18.8.10 4-conductor polarized connector with male contacts



18.9 Test blocks

18.9.1 Female portion with short-circuiting bar (with terminals shown)



18.9.2 Male portion (with terminals shown)



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

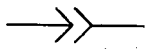
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Check the source to verify that this is the current version before use.

LIST OF SYMBOLS

18.10 WAVEGUIDE FLANGES

Note 18.10A—If necessary to indicate flange type (plain or choke), place waveguide recognition symbol immediately adjacent to the connector symbol. If the recognition symbol is omitted or is spaced from the connector symbol, the nature of the flange is not to be implied from the symbol.

18.10.1 Mated (General)


See note 18.10A

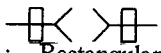
18.10.2 Plain (Rectangular waveguide)


See note 18.10A

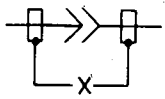
18.10.3 Choke (Rectangular waveguide)


See note 18.10A

18.10.3.1. Application: Mated choke flanges in rectangular waveguide line



18.10.4 Application: Rectangular waveguide with mated plain and choke flanges with DC isolation (insulation) between sections of waveguide.



19 RESERVED

20 RESERVED

21 RESERVED

22 RESERVED

23 CONTACT, ELECTRICAL F

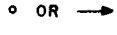
For build-ups or forms using electrical contacts, see applications under CONNECTOR (item 18), RELAY (item 66), SWITCH (item 76), and SWITCHING FUNCTION (item 77). See DRAFTING PRACTICES (par. 3.8).

23.1 Fixed contact

23.1.1. Fixed contact for jack, key, relay, etc.



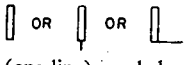
23.1.2 Fixed contact for switch



23.1.3 Fixed contact for momentary switch
See SWITCH (items 76.8 and 76.10).

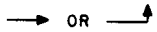


23.1.4 Sleeve



23.2 Moving contact

23.2.1 Adjustable or sliding contact for resistor, inductor, etc.

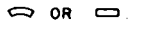


23.2.2 Locking



23.2.3 Nonlocking

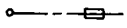
23.2.4 Segment; bridging contact
See SWITCH (items 76.12.3 and 76.12.4).



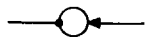
23.2.5 Vibrator reed



23.2.6 Vibrator split reed



23.2.7 Rotating contact (slip ring) and brush



23.3 Basic contact assemblies

The standard method of showing a contact is by a symbol indicating the circuit condition it produces when the actuating device is in the de-energized or nonoperated position. The actuating device may be of a mechanical, electrical, or other nature, and a clarifying note may be necessary with the symbol to explain the proper point at which the contact functions, for example, the point where a contact closes or opens as a function of changing pressure, level, flow, voltage, current, etc. In cases where it is desirable to show contacts in the energized or operated condition and where confusion may result, a clarifying note shall be added to the drawing.

Auxiliary switches or contacts for circuit breakers, etc., may be designated as follows:

(*a*) Closed when device is in energized or operated position,

(*b*) Closed when device is in de-energized or nonoperated position,

(*aa*) Closed when operating mechanism of main device is in energized or operated position,

(*bb*) Closed when operating mechanism of main device is in de-energized or nonoperated position.

See American Standard C37.2 for further details.

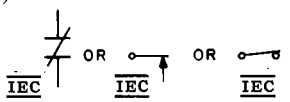
In the parallel-line contact symbols shown below, the length of the parallel lines shall be approximately 1¼ times the width of the gap (except for item 23.6)

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

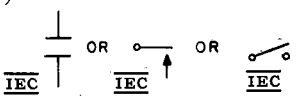
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LIST OF SYMBOLS

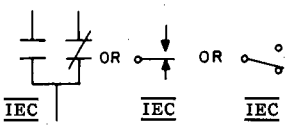
23.3.1 Closed contact (break)
See also SWITCHING FUNCTION (item 77).



23.3.2 Open contact (make)
See also SWITCHING FUNCTION (item 77).



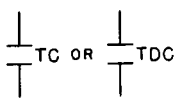
23.3.3 Transfer
See also SWITCHING FUNCTION (item 77).



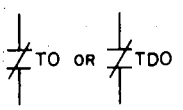
23.3.4 Make-before-break



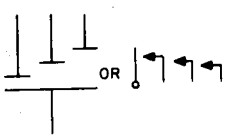
23.4 Application: open contact with time closing (TC) or time delay closing (TDC) feature



23.5 Application: closed contact with time opening (TO) or time delay opening (TDO) feature



23.6 Time sequential closing



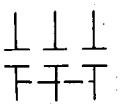
24 CONTACTOR

See also RELAY (item 66).

Fundamental symbols for contacts, coils, mechanical connections, etc., are the basis of contactor symbols and should be used to represent contactors on complete diagrams. Complete diagrams of contactors consist of combinations of fundamental symbols for control coils, mechanical connections, etc., in such configurations as to represent the actual device.

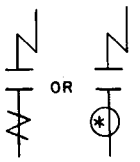
Mechanical interlocking should be indicated by notes.

24.1 Manually operated 3-pole contactor



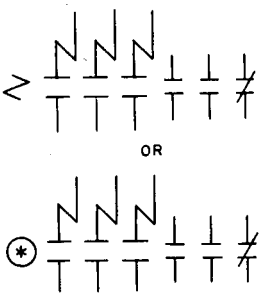
24.2 Electrically operated 1-pole contactor with series blowout coil

Note 24.2A—The asterisk is not a part of the symbol. Always replace the asterisk by a device designation.



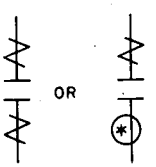
* See note 24.2A

24.3 Electrically operated 3-pole contactor with series blowout coils; 2 open and 1 closed auxiliary contacts (shown smaller than the main contacts)



* See note 24.2A

24.4 Electrically operated 1-pole contactor with shunt blowout coil



* See note 24.2A

25 CORE

25.1 General or air core
If it is necessary to identify an air core, a note should appear adjacent to the symbol of the inductor or transformer.

NO SYMBOL

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

25.2 Magnetic core of inductor or transformer

Not to be used unless it is necessary to identify a magnetic core.

See INDUCTOR (item 42.2) and TRANSFORMER (item 86.2).



25.3 Core of magnet

For use if representation of the core is necessary.

See MAGNET, PERMANENT (item 47).



26 COUNTER, ELECTROMAGNETICALLY OPERATED MESSAGE REGISTER

26.1 General



26.2 With a make contact



27 COUPLER, DIRECTIONAL F

Commonly used in coaxial and waveguide diagrams.

The arrows indicate the direction of power flow.

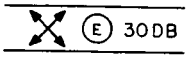
Number of coupling paths, type of coupling, and transmission loss may be indicated.

27.1 General

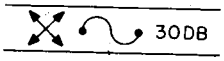


27.2 Applications

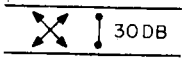
27.2.1 *E*-plane aperture coupling, 30-decibel transmission loss



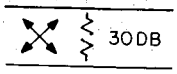
27.2.2 Loop coupling, 30-decibel transmission loss



27.2.3 Probe coupling, 30-decibel transmission loss



27.2.4 Resistance coupling, 30-decibel transmission loss



28 COUPLING

Commonly used in coaxial and waveguide diagrams.

28.1 Coupling by aperture with an opening of less than full waveguide size

Transmission loss may be indicated.

Note 28.1A—The asterisk is not a part of the symbol. Always replace the asterisk by E, H, or HE, depending on the type of coupling.

E indicates that the physical plane of the aperture is perpendicular to the transverse component of the major *E* lines.

H indicates that the physical plane of the aperture is parallel to the transverse component of the major *E* lines.

HE indicates coupling by all other kinds of apertures.

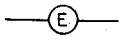


* See note 28.1A

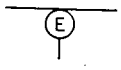
28.1.1 Application: *E*-plane coupling by aperture to space



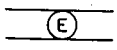
28.1.2 Application: *E*-plane coupling by aperture; 2 ends of transmission path available



28.1.3 Application: *E*-plane coupling by aperture; 3 ends of transmission path available



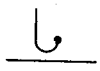
28.1.4 Application: *E*-plane coupling by aperture; 4 ends of transmission path available



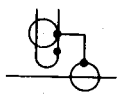
28.2 Coupling by loop to space



28.2.1 Coupling by loop to guided transmission path



28.2.2 Application: coupling by loop from coaxial to circular waveguide with direct-current grounds connected



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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28.3 Coupling by probe to space

See OPEN CIRCUIT (item 82.2).

28.3.1 Application: coupling by probe to a guided transmission path



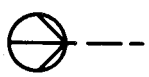
28.3.2 Application: coupling by probe from coaxial to rectangular waveguide with direct-current grounds connected



29 RESERVED

30 ‡DETECTOR, PRIMARY; MEASURING TRANSDUCER

See also HALL GENERATOR (item 39) and THERMAL CONVERTER (item 48)

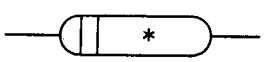


31 DELAY FUNCTION DELAY LINE F SLOW-WAVE STRUCTURE

31.1 General

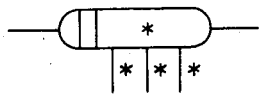
Note 31.1A—Length of delay may be indicated. Asterisk is not part of symbol. See also note 12B.

Note 31.1B—The two vertical lines indicate input side.



* See note 31.1A

31.2 Tapped delay function



* See note 31.1A

31.3 Bifilar slow-wave structure

Commonly used in traveling-wave tubes.

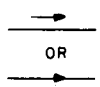


* See note 31.1A

32 DIRECTION OF FLOW OF POWER, SIGNAL, OR INFORMATION

32.1 One-way

Note 32.1A—The lower symbol is used if it is necessary to conserve space. The arrowhead in the lower symbol shall be filled.



See note 32.1A

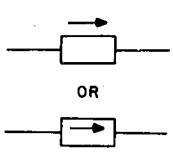
32.2 Both ways



See note 32.1A

32.3 Application: one-way circuit element, general

Note 32.3A—In all cases indicate the type of apparatus by appropriate words or letters in the rectangle.



See note 32.3A

‡ *The broken line - — - indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

33 DISCONTINUITY

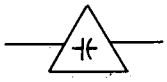
A component that exhibits throughout the frequency range of interest the properties of the type of circuit element indicated by the symbol within the triangle.

Commonly used for coaxial and waveguide transmission.

33.1 Equivalent series element, general



33.1.1 Capacitive reactance



33.1.2 Inductive reactance



33.1.3 Inductance-capacitance circuit with infinite reactance at resonance



33.1.4 Inductance-capacitance circuit with zero reactance at resonance



33.1.5 Resistance



33.2 Equivalent shunt element, general



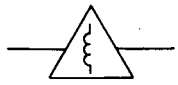
33.2.1 Capacitive susceptance



33.2.2 Conductance



33.2.3 Inductive susceptance



33.2.4 Inductance-capacitance circuit with infinite susceptance at resonance



33.2.5 Inductance-capacitance circuit with zero susceptance at resonance



34 ELECTRON TUBE E

See also ENVELOPE; ENCLOSURE (item 35); and TERMINALS (item 81); (for electron tubes, semiconductor devices, etc.)

Tube-component symbols are shown first. These are followed by typical applications showing the use of these specific symbols in the various classes of devices such as thermionic, cold-cathode, and photoemissive tubes of varying structures and combinations of elements (triodes, pentodes, cathode-ray tubes, magnetrons, etc.).

Lines outside of the envelope are not part of the symbol but are electrical connections thereto.

Connections between the external circuit and electron tube symbols within the envelope may be located as required to simplify the diagram.

34.1 Emitting electrode

34.1.1 Directly heated (filamentary) cathode

Note 34.1.1A—Leads may be connected in any convenient manner to ends of the ⌒ provided the identity of the ⌒ is retained.

⌒ IEC

See note 34.1.1A

34.1.1.1 With tap

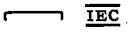
See item 34.10.3.

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

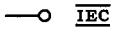
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LIST OF SYMBOLS

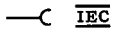
34.1.2 Indirectly heated cathode
Lead may be connected to either extreme end of the ⌐¬ or, if required, to both ends, in any convenient manner.



34.1.3 Cold cathode (including ionically heated cathode)



34.1.4 Photocathode



34.1.5 Pool cathode

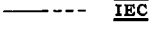


34.1.6 Ionically heated cathode with provision for supplementary heating
See note 34.1.1A.

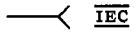


34.2 Controlling electrode

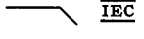
34.2.1 Grid (including beam-confining or beam-forming electrodes)



34.2.2 Deflecting electrodes (used in pairs); reflecting or repelling electrode (used in velocity-modulated tube)



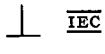
34.2.3 Ignitor (in pool tubes) (should extend into pool); Starter (in gas tubes)



34.2.4 Excitor (contactor type)

34.3 Collecting electrode

34.3.1 Anode or plate (including collector electrode and fluorescent target)



34.3.2 Target or X-ray anode
Drawn at about a 45-degree angle.



34.4 Collecting and emitting electrode

34.4.1 Dynode



34.4.2 Alternately collecting and emitting electrode

34.4.2.1 Composite anode-photocathode



34.4.2.2 Composite anode-cold cathode



34.4.2.3 Composite anode-ionically heated cathode with provision for supplementary heating
See note 34.1.1A.



34.5 Heater
See note 34.1.1A.

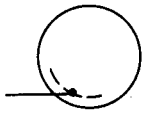


34.5.1 With tap
See item 34.10.3.

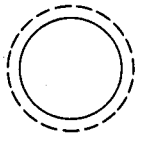
34.6 Envelope
See item 35.

34.7 Shield
See item 34.10.10.
This is understood to shield against electric fields unless otherwise noted.

34.7.1 Any shield against electric fields that is within the envelope and that is connected to an independent terminal



34.7.2 Outside envelope of X-ray tube



34.8 Coupling
See COUPLING (item 28) and PATH, TRANSMISSION (items 58.8.2 and 58.11).

34.8.1 Coupling by loop (electromagnetic type)
Coupling loop may be shown inside or outside envelope as desired.



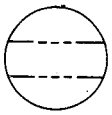
Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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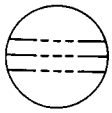
LIST OF SYMBOLS

34.9 Resonators (cavity type)

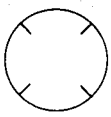
34.9.1 Single-cavity envelope and grid-type associated electrodes



34.9.2 Double-cavity envelope and grid-type associated electrodes



34.9.3 Multicavity magnetron anode and envelope



34.10 General notes

34.10.1 If new symbols are necessary, they should be formed where possible from component symbols. For example, see DYNODE (item 34.4.1), which combines the anode and photocathode convention.

34.10.2 A connection to anode, dynode, pool cathode, photocathode, deflecting electrode, composite anode-photocathode, and composite anode-cold cathode shall be to the center of that symbol. Connection to any other electrode may be shown at either end or both ends of the electrode symbol.

34.10.3 A diagram for a tube having more than one heater or filament shall show only one heater or filament symbol ⌒ unless they have entirely separate connections. If a heater or filament tap is made, either brought out to a terminal or internally connected to another element, it shall be connected at the vertex of the symbol, regardless of the actual division of voltage across the heater or filament.

34.10.4 Standard symbols, such as the inclined arrow for tunability and connecting dotted lines for ganged components, may be added to a tube symbol to extend the meaning of the tube symbol, provided such added feature or component is integral with the tube.

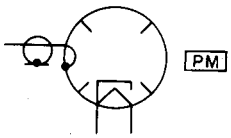
34.10.5 Electric components, such as resistors, capacitors, or inductors, which are integral parts of the tube and are important to its functional operation, shall be shown in the standard manner.

34.10.6 Multiple equipotential cathodes that are directly connected inside the tube shall be shown as a single cathode.

34.10.7 A tube having two or more grids tied internally shall be shown with symbols for each grid, except when the grids are adjacent in the tube structure. Thus, the diagram for a twin pentode having a common screen-grid connection for each section and for a converter tube having the No. 3 and No. 5 grids connected internally will show separate symbols for each grid. However, a triode where the control grid is physically in the form of two grid windings would show only one grid.

34.10.8 A tube having a grid adjacent to a plate but internally connected to the plate to form a portion of it shall be shown as having a plate only.

34.10.9 Associated parts of a circuit, such as focusing coils, deflecting coils, field coils, etc., are not a part of the tube symbol but may be added to the circuit in the form of standard symbols. For example, resonant-type magnetron with permanent magnet may be shown:

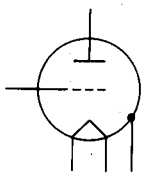


34.10.10 External and internal shields, whether integral parts of tubes or not, shall be omitted from the circuit diagram unless the circuit diagram requires their inclusion.

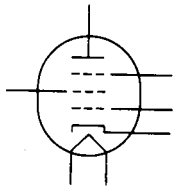
34.10.11 In line with standard drafting practice, straight-line crossovers are recommended.

34.11 Typical applications

34.11.1 Triode with directly heated filamentary cathode and envelope connection to base terminal



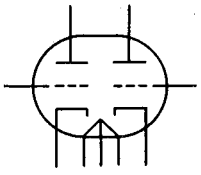
34.11.2 Equipotential-cathode pentode showing use of elongated envelope



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

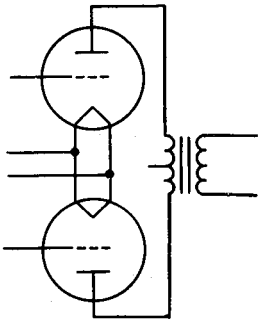
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34.11.3 Equipotential-cathode twin triode illustrating elongated envelope and rule of item 34.10.3



34.11.4 Typical wiring figure

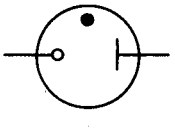
This figure illustrates how tube symbols may be placed in any convenient position in a circuit.



34.11.5 Cold-cathode gas-filled tube

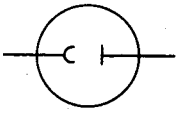
34.11.5.1 Rectifier; voltage regulator for direct-current operation

See also LAMP, GLOW (item 44.3).

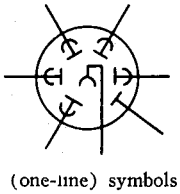


34.11.6 Phototube

34.11.6.1 Single-unit, vacuum type

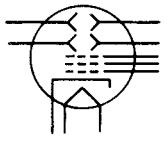


34.11.6.2 Multiplier type

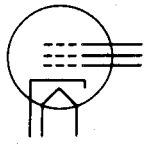


34.11.7 Cathode-ray tube

34.11.7.1 With electric-field deflection



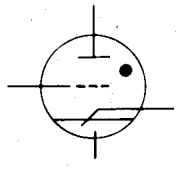
34.11.7.2 For magnetic deflection



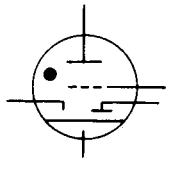
34.11.8 Mercury-pool tube

See also RECTIFIER (item 65.4).

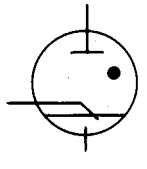
34.11.8.1 With ignitor and control grid



34.11.8.2 With excitor, control grid, and holding anode

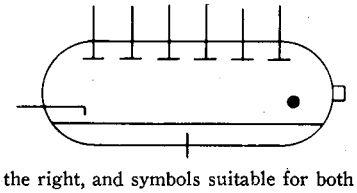


34.11.8.3 Single-anode pool-type vapor rectifier with ignitor



34.11.8.4 6-anode metallic-tank pool-type vapor rectifier with excitor, showing rigid-terminal symbol for control connection to tank (pool cathode is insulated from tank)

Anode symbols are located as convenient.



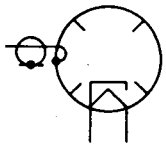
Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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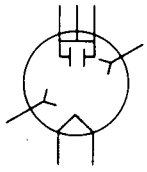
LIST OF SYMBOLS

34.11.9 Magnetron

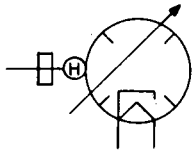
34.11.9.1 Resonant type with coaxial output



34.11.9.2 Transit-time split-plate type with stabilizing deflecting electrodes and internal circuit

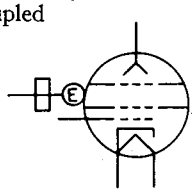


34.11.9.3 Tunable, aperture coupled

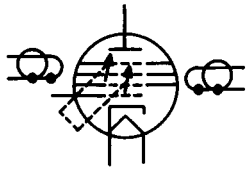


34.11.10 Velocity-modulation (velocity-variation) tube

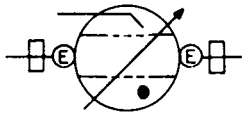
34.11.10.1 Reflex klystron, integral cavity, aperture coupled



34.11.10.2 Double-cavity klystron, integral cavity, permanent external-ganged tuning, loop coupled (coupling loop may be shown inside if desired) See item 34.8.1.

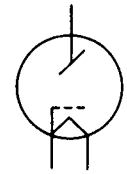


34.11.11 Transmit-receive (TR) tube
Gas filled, tunable integral cavity, aperture coupled, with starter.

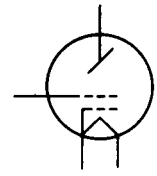


34.11.12 X-ray tube

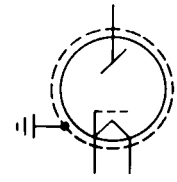
34.11.12.1 With filamentary cathode and focusing grid (cup). The anode may be cooled by fluid or radiation.



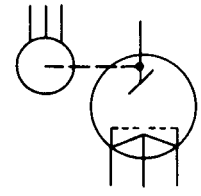
34.11.12.2 With control grid, filamentary cathode, and focusing cup



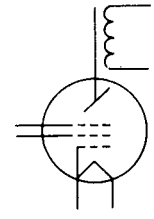
34.11.12.3 With grounded electrostatic shield



34.11.12.4 Double focus with rotating anode (see item 34.10.9)



34.11.12.5 With multiple accelerating electrode, electrostatically and electromagnetically focused
See item 34.10.9.

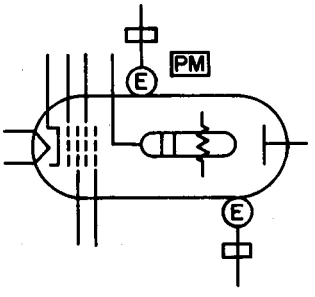


Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

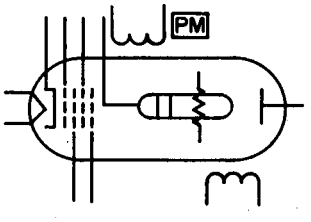
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34.11.13 Traveling-wave-tube

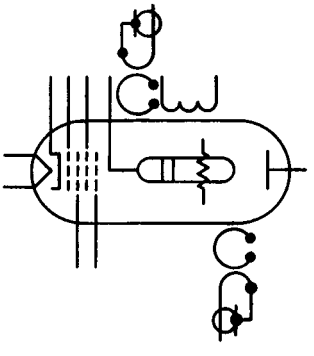
34.11.13.1 Forward-wave traveling-wave-tube amplifier shown with four grids, having slow-wave structure with attenuation, magnetic focusing by external permanent magnet, rf input and rf output coupling each by E-plane aperture to external rectangular waveguide



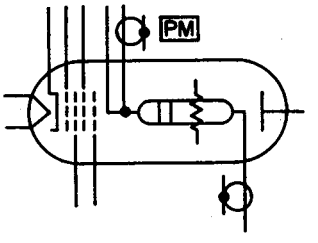
34.11.13.2 Forward-wave traveling-wave-tube amplifier shown with four grids, having slow-wave structure with attenuation, magnetic focusing by external permanent magnet, rf input and rf output coupling each by inductive coupling



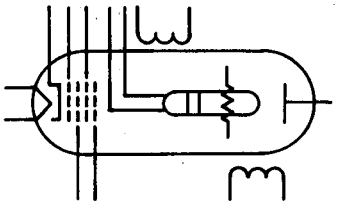
34.11.13.3 Forward-wave traveling-wave-tube amplifier shown with four grids, having slow-wave structure with attenuation, external electromagnetic focusing, rf input and rf output coupling each by external cavity and loop coupling to a coaxial path



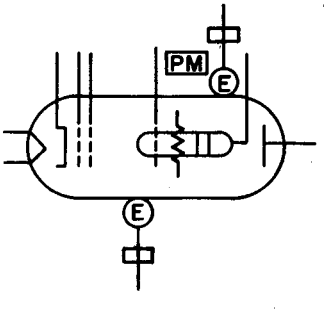
34.11.13.4 Forward-wave traveling-wave-tube amplifier shown with four grids, having slow-wave structure with attenuation, magnetic focusing by external permanent magnet, rf input and rf output coupling each by direct connection from slow-wave structure to a coaxial path



34.11.13.5 Forward-wave traveling-wave-tube amplifier shown with four grids, having bifilar slow-wave structure with attenuation, electrostatic focusing, rf input and rf output coupling each by inductive coupling



34.11.13.6 Backward-wave traveling-wave-tube amplifier shown with two grids, having slow-wave structure with attenuation, sole (beam-aligning electrode), magnetic focusing by external permanent magnet, rf input and rf output coupling each by E-plane aperture to external rectangular waveguide

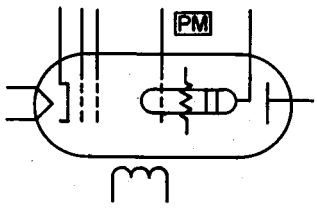


Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

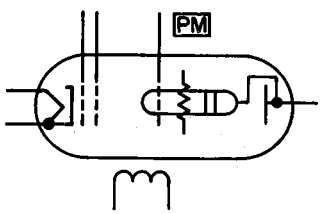
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
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LIST OF SYMBOLS

34.11.13.7 Backward-wave traveling-wave-tube oscillator shown with two grids, having slow-wave structure with attenuation, sole (beam-aligning electrode), magnetic focusing by external permanent magnet, rf output coupling by inductive coupling



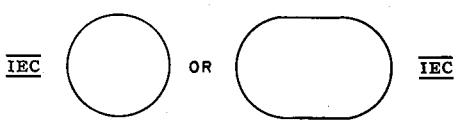
34.11.13.8 Backward-wave traveling-wave-tube oscillator shown with two grids, having slow-wave structure with attenuation; sole (beam-aligning electrode), magnetic focusing by external permanent magnet, rf output coupling by inductive coupling, with slow-wave structure connected internally to collector



35 ENVELOPE ENCLOSURE

The general envelope symbol identifies the envelope or enclosure regardless of evacuation or pressure. When used with electron-tube component symbols, the general envelope symbol indicates a vacuum enclosure unless otherwise specified. A gas-filled device may be indicated by a dot within the envelope symbol.

35.1 General



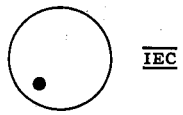
35.1.1 Split envelope

If necessary, envelope may be split.

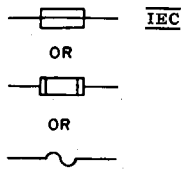


35.2 Gas-filled envelope

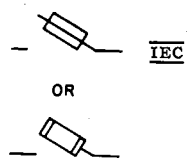
The dot may be located as convenient.



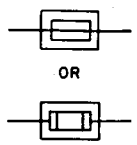
36 FUSE



36.1 High-voltage primary fuse cutout, dry

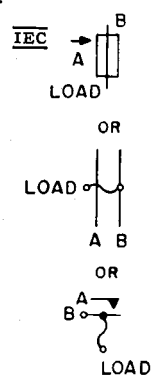


36.2 High-voltage primary fuse cutout, oil



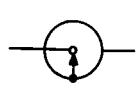
36.3 With alarm contact

When fuse blows, alarm bus A is connected to power bus B. Letters are for explanation and are not part of the symbol.



37 GOVERNOR (Contact-making) SPEED REGULATOR

Contacts open or closed as required (shown here as closed).



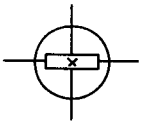
Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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38 GROUND

See CIRCUIT RETURN (item 13.1).

39 HALL GENERATOR

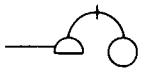


40 HANDSET
OPERATOR'S SET

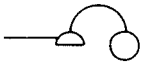
40.1 General



40.2 With push-to-talk switch

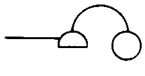


40.3 3-conductor handset



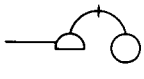


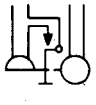
40.4 4-conductor handset





40.5 4-conductor handset with push-to-talk switch





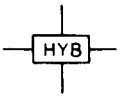
40.6 Operator's set





41 HYBRID
DIRECTIONALLY SELECTIVE
TRANSMISSION DEVICES

41.1 Hybrid, general

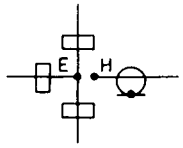


41.2 Hybrid, junction

Commonly used in coaxial and waveguide transmission.



41.3 Application: rectangular waveguide and coaxial coupling



41.4 Hybrid, circular (basic)

Note 41.4A—The asterisk is not a part of the symbol. Always replace the asterisk by E, H, or HE. E indicates that there is a principal E transverse field in the plane of the ring. H indicates that there is a principal H transverse field in the plane of the ring. HE shall be used for all other cases.

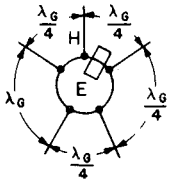
An arm that has coupling of a different type from that designated above shall be marked according to COUPLING (item 28.1).

Critical distances should be labeled in terms of guide wavelengths.



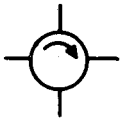
* See note 41.4A.

41.4.1 Application: 5-arm circular hybrid with principal coupling in the *E* plane and with 1-arm *H* coupling using rectangular waveguide



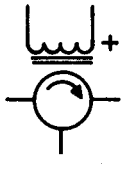
41.5 Circulator, fixed direction

Arrowhead indicates direction of power flow from any input to next adjacent arm but not to any other arm. Circulator may have three or more ports.



41.5.1 Application: circulator, reversible direction

The polarity symbol (item 63.1) must be used with the electromagnet symbol to indicate proper direction flow.



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

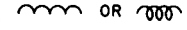
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LIST OF SYMBOLS

42 INDUCTOR
WINDING F (Machine or transformer)
REACTOR F

See also COIL, OPERATING (item 16) and TERMINATION (item 82).

42.1 General
Either symbol may be used in the following items.

IEC  OR

42.2 If it is desired especially to distinguish magnetic-core inductors



42.3 Tapped



42.4 Application: adjustable inductor



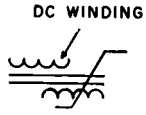
42.5 Application: adjustable or continuously adjustable inductor

 IEC

42.6 Shunt inductor



42.7 Saturable-core inductor (reactor)
Polarity marks may be added to direct-current winding.
Explanatory words and arrow are not part of the symbol shown.



42.8 Application: carrier line trap
Note 42.8A—Asterisk is not part of the symbol. The symbol represents a general trap or single-frequency trap unless one of the following letter combinations is used.

2f Two-frequency
WB Wide band



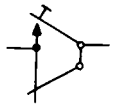
* See note 42.8A

43 KEY, TELEGRAPH F

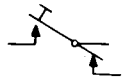
43.1 Simple



43.2 Simple with shorting switch

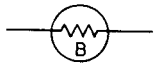


43.3 Open circuit or pole changing



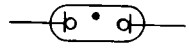
44 LAMP

44.1 Ballast lamp; ballast tube
The primary characteristic of the element within the circle is designed to vary nonlinearly with the temperature of the element.



44.2 Lamp, fluorescent F

44.2.1 2-terminal

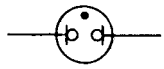


44.2.2 4-terminal



44.3 Lamp, glow F, cold-cathode lamp; neon lamp

44.3.1 Alternating-current type



44.3.2 Direct-current type
See also ELECTRON TUBE (item 34.11.5.1).



44.4 Lamp, incandescent F (Incandescent-filament illuminating lamp)



44.5 Indicating lamp; switchboard lamp
See VISUAL SIGNALING DEVICE (item 88).

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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45 LIMITER, CURRENT (For power cable)
The arrowheads in this case are filled.

Note 45A—Use appropriate number of single-line diagram symbols.



See note 45A

46 MACHINE, ROTATING

46.1 Basic



46.2 Generator (general)



46.3 Motor (general)



46.4 Motor, multispeed

USE BASIC MOTOR SYMBOL AND NOTE SPEEDS

46.5 ‡ Rotating armature with commutator and brushes

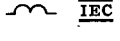




46.6 Field, generator or motor

Either symbol of item 42.1 may be used in the following items.

46.6.1 Compensating or commutating



46.6.2 Series



46.6.3 Shunt, or separately excited



46.6.4 Magnet, permanent

See item 47.

46.7 Winding symbols

Motor and generator winding symbols may be shown in the basic circle using the following representations.

46.7.1 1-phase



46.7.2 2-phase



46.7.3 3-phase wye (ungrounded)



46.7.4 3-phase wye (grounded)



46.7.5 3-phase delta



46.7.6 6-phase diametrical

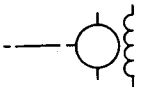


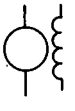
46.7.7 6-phase double-delta



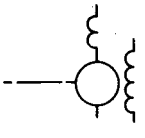
46.8 Direct-current machines; applications

46.8.1. ‡ Separately excited direct-current generator or motor



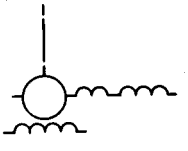


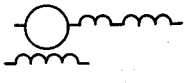
46.8.2 ‡ Separately excited direct-current generator or motor; with commutating or compensating field winding or both



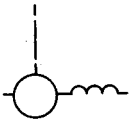


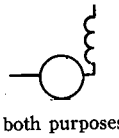
46.8.3 ‡ Compositely excited direct-current generator or motor; with commutating or compensating field winding or both





46.8.4 ‡ Direct-current series motor or 2-wire generator





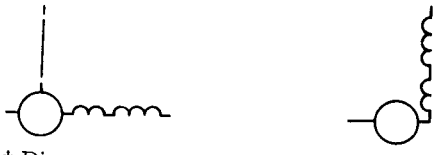
‡ *The broken line - - - indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

46.8.5 ‡ Direct-current series motor or 2-wire generator; with commutating or compensating field winding or both



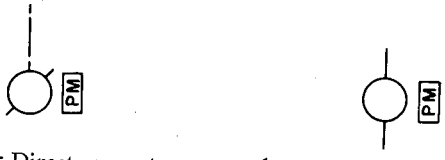
46.8.6 ‡ Direct-current shunt motor or 2-wire generator



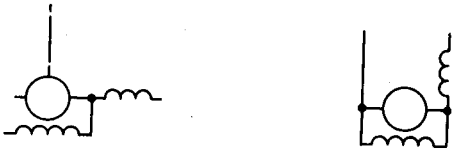
46.8.7 ‡ Direct-current shunt motor or 2-wire generator; with commutating or compensating field winding or both



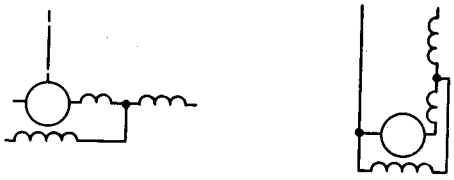
46.8.8 ‡ Direct-current permanent-magnet-field generator or motor



46.8.9 ‡ Direct-current compound motor or 2-wire generator or stabilized shunt motor



46.8.10 ‡ Direct-current compound motor or 2-wire generator or stabilized shunt motor; wih commutating or compensating field winding or both



46.8.11 ‡ Direct-current 3-wire shunt generator



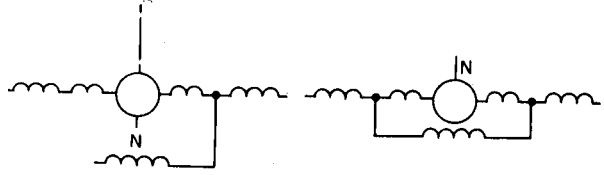
46.8.12 ‡ Direct-current 3-wire shunt generator; with commutating or compensating field winding or both



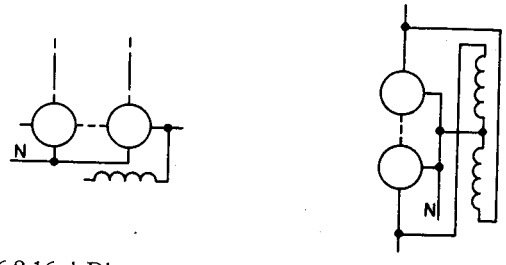
46.8.13 ‡ Direct-current 3-wire compound generator



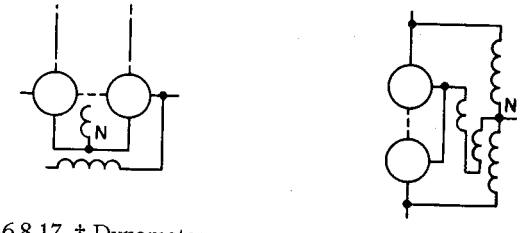
46.8.14 ‡ Direct-current 3-wire compound generator; with commutating or compensating field winding or both



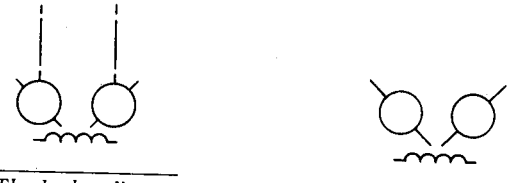
46.8 15 ‡ Direct-current balancer, shunt wound



46.8.16 ‡ Direct-current balancer, compound wound



46.8.17 ‡ Dynamotor



‡ *The broken line - - - indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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46.8.18 ‡ Double-current generator



46.8.19 ‡ Acyclic generator, separately excited



46.8.20 ‡ Regulating generator (rotary amplifier) shunt wound with short-circuited brushes



46.8.21 ‡ Regulating generator (rotary amplifier) shunt wound without short-circuited brushes



46.8.22 ‡ Regulating generator (rotary amplifier) shunt wound with compensating field winding and short-circuited brushes



46.8.23 ‡ Regulating generator (rotary amplifier) shunt wound with compensating field winding but without short-circuited brushes



46.9 Alternating-current machines; applications

46.9.1 ‡ Squirrel-cage induction motor or generator, split-phase induction motor or generator, rotary phase converter, or repulsion motor



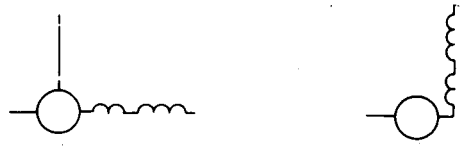
46.9.2 ‡ Wound-rotor induction motor, synchronous induction motor, induction generator, or induction frequency converter



46.9.3 ‡ Alternating-current series motor



46.9.4 ‡ Alternating-current series motor; with commutating or compensating field winding or both



46.9.5 ‡ 1-phase shaded-pole motor



46.9.6 ‡ 1-phase repulsion-start induction motor



46.9.7 ‡ 1-phase hysteresis motor



46.9.8 ‡ Reluctance motor



46.9.9 ‡ 1-phase subsynchronous reluctance motor



46.9.10 ‡ Magnetoelectric generator, 1 phase



46.9.11 ‡ Shunt-characteristic brush-shifting motor



‡ *The broken line - - - indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

46.9.12 ‡ Series-characteristic brush-shifting motor with 3-phase rotor



46.9.13 Series-characteristic brush-shifting motor with 6- or 8-phase rotor



46.9.14 Ohmic-drop exciter with 3- or 6-phase input



46.9.15 Ohmic-drop exciter with 3- or 6-phase input, with output leads



46.9.16 3-phase regulating machine



46.9.17 Phase shifter with 1-phase output

See PHASE SHIFTER (item 60).

See TRANSFORMER (item 86).



46.9.18 Phase shifter with 3-phase output

See PHASE SHIFTER (item 60).

See TRANSFORMER (item 86).



46.10 Alternating-current machines with direct-current field excitation; applications

46.10.1 ‡ Synchronous motor, generator, or condenser



46.10.2 ‡ Synchronous motor, generator, or condenser with neutral brought out



46.10.3 ‡ Synchronous motor, generator, or condenser with both ends of each phase brought out



46.10.4 ‡ Double-winding synchronous generator, motor, or condenser



46.10.5 ‡ Synchronous-synchronous frequency changer



46.10.6 ‡ Synchronous induction frequency changer



46.11 Alternating- and direct-current composite machines; applications

46.11.1 ‡ Synchronous or regulating-pole converter



‡ *The broken line ---- indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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Check the source to verify that this is the current version before use.

LIST OF SYMBOLS

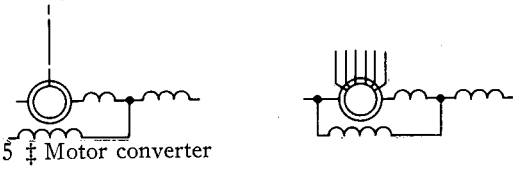
46.11.2 ‡ Synchronous booster or regulating-pole converter; with commutating or compensating field windings or both



46.11.3 ‡ Synchronous shunt-wound converter; with commutating or compensating windings or both



46.11.4 ‡ Synchronous converter compound wound: with commutating or compensating field windings or both



46.11.5 ‡ Motor converter



47 MAGNET, PERMANENT F

PM

48 METER
INSTRUMENT

Note 48A—The asterisk is not a part of the symbol. Always replace the asterisk by one of the following letter combinations, depending on the function of the meter or instrument, unless some other identification is provided in the circle and explained on the diagram.

A	Ammeter F IEC
AH	Ampere-hour
CMA	Contact-making (or breaking) ammeter
CMC	Contact-making (or breaking) clock
CMV	Contact-making (or breaking) voltmeter
CRO	Oscilloscope or cathode-ray oscillograph
DB	DB (decibel) meter
DBM	DBM (decibels referred to 1 milliwatt) meter
DM	Demand meter
DTR	Demand-totalizing relay
F	Frequency meter F
G	Galvanometer F
GD	Ground detector
I	Indicating
INT	Integrating
μA or UA	Microammeter
MA	Milliammeter
NM	Noise meter
OHM	Ohmmeter F
OP	Oil pressure
OSCG	Oscillograph, string
PH	Phasemeter F
PI	Position indicator
PF	Power Factor
RD	Recording demand meter
REC	Recording
RF	Reactive factor
SY	Synchroscope
TLM	Telemeter
T	Temperature meter
THC	Thermal converter
TT	Total time
V	Voltmeter F IEC
VA	Volt-ammeter
VAR	Varmeter F
VARH	Varhour meter
VI	Volume indicator; Meter, audio level F
VU	Standard volume indicator; Meter, audio level F
W	Wattmeter F IEC
WH	Watthour meter

(*)

* See note 48A

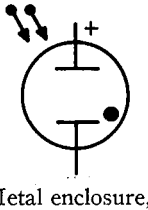
49 RESERVED

50 NUCLEAR-RADIATION DETECTOR, gas filled

IONIZATION CHAMBER
PROPORTIONAL COUNTER TUBE
GEIGER-MÜLLER COUNTER TUBE

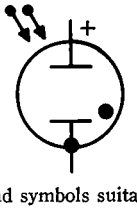
Note 50A—For other types of radiation-sensitivity indicators, see item 69.

50.1 General



See note 50A

50.2 Application: Metal enclosure, having one collector connected to the enclosure



See note 50A

‡ *The broken line - - - indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

51 MICROPHONE F




52 MODE SUPPRESSION F

Commonly used in coaxial and waveguide transmission.



53 MODE TRANSDUCER F

Commonly used in coaxial and waveguide diagrams.

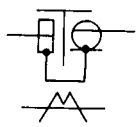
53.1 General



53.2 Application: transducer from rectangular to circular waveguide



53.3 Application: transducer from rectangular waveguide to coaxial with mode suppression and direct-current grounds connected



54 MOTION, MECHANICAL

54.1 Translation, one direction



54.2 Translation, both directions



54.3 Rotation, one direction



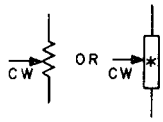
54.4 Rotation, both directions



54.5 Rotation designation (applied to a resistor)

CW indicates position of adjustable contact at the limit of clockwise travel viewed from knob or actuator end unless otherwise indicated.

Note 54.5A—The asterisk is not a part of the symbol. Always add identification within or adjacent to the rectangle.



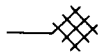
* See note 54.5A

55 NETWORK

55.1. General



55.2 Network, low-voltage power



56 OSCILLATOR
GENERALIZED ALTERNATING-CURRENT SOURCE



57 PAD (ATTENUATOR, FIXED F)

See also ATTENUATOR (item 5).

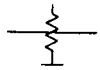
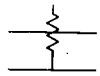
57.1 General



57.2 Balanced, general

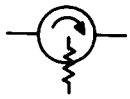



57.3 Unbalanced, general

57.4 Unidirectional (Isolator)

Power flowing in direction of arrow is not intentionally attenuated.



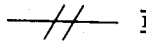
58 PATH, TRANSMISSION
CONDUCTOR
CABLE
WIRING

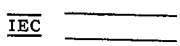
58.1 Guided path, general

A single line represents the entire group of conductors or the transmission path needed to guide the power or the signal. For coaxial and waveguide work, the recognition symbol is used at the beginning and end of each kind of transmission path and at intermediate points as needed for clarity. In waveguide work, mode may be indicated.

58.2 Conductive path or conductor; wire

58.2.1 Two conductors or conductive paths



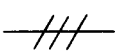


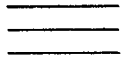
Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

58.2.2 Three conductors or conductive paths

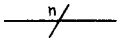




58.2.3 "*n*" conductors or conductive paths

Note 58.2.3A—The "*n*" is not a part of the symbol. A number representing the actual number of paths shall be substituted for "*n*"

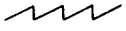
Note 58.2.3B—Draw individual paths.



See note 58.2.3B

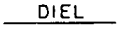
See note 58.2.3A

58.3 Air or space path



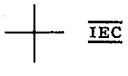
58.4 Dielectric path other than air

Commonly used for coaxial and waveguide transmission.



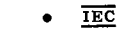
58.5 Crossing of paths or conductors not connected

The crossing is not necessarily at a 90-degree angle.

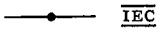


58.6 Junction of paths or conductors

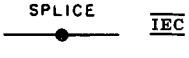
58.6.1 Junction (if desired)



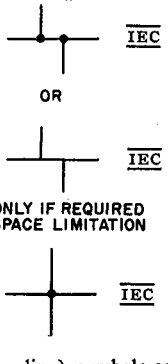
58.6.1.1 Application: junction of paths, conductor, or cable. If desired indicate path type, or size



58.6.1.2 Application: splice (if desired) of same size cables. Junction of conductors of same size or different size cables. If desired indicate sizes of conductors

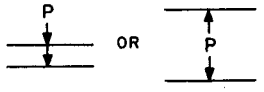


58.6.2 Junction of connected paths, conductors, or wires

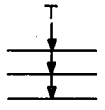


58.7 Associated conductors

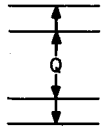
58.7.1 Pair (twisted unless otherwise specified)



58.7.2 Triple (twisted unless otherwise specified)



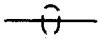
58.7.3 Quad



58.8 Assembled conductors; cable

Commonly used in communication diagrams.

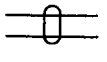
58.8.1 Shielded single conductor



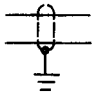
58.8.2 Coaxial cable, recognition symbol
Coaxial transmission path
Cable, radio frequency E, (Coaxial)
See item 58.1.



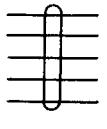
58.8.3 2-conductor cable



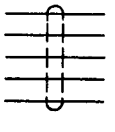
58.8.4 Shielded 2-conductor cable with shield grounded



58.8.5 5-conductor cable



58.8.6 Shielded 5-conductor cable

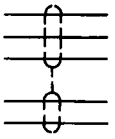


Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

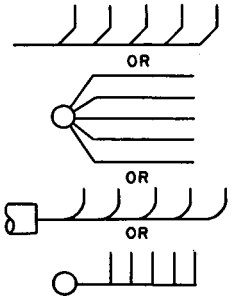
58.8.6.1 Shielded 5-conductor cable with conductors separated on the diagram for convenience



58.8.7 Cable underground or in conduit (*long dashes*)

58.8.8 Grouping of leads

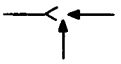
Normally, bend of line indicates direction of conductor joining cable.



58.9 Alternate or conditional wiring

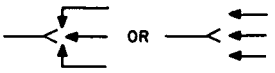
Not commonly used on power diagrams.
The arrowheads in this case shall be solid.

Note 58.9A—A note shall explain the connections.



See note 58.9A

58.9.1 Application: 3 alternate paths



See note 58.9A

58.10 Associated or future (*short dashes*)

58.10.1 Application: associated or future equipment (amplifier shown)



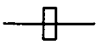
58.11 Waveguide E

See item 58.1. If necessary, indicate special characteristics by a note.

58.11.1 Circular, recognition symbol

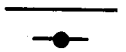


58.11.2 Rectangular, recognition symbol

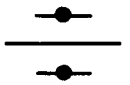


58.12 Strip-type transmission line

58.12.1 Strip line



58.12.2 Sandwich line

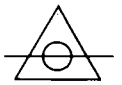


59 RESERVED

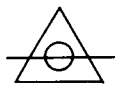
60 PHASE SHIFTER NETWORK, PHASE CHANGING E

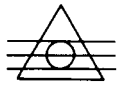
For power circuits see MACHINE, ROTATING (items 46.9.17 and 46.9.18).

60.1 General



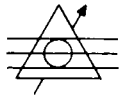
60.2 3-wire or 3-phase





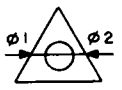
60.2.1 Application: adjustable





60.3 Differential phase shifter

Phase shift ϕ in direction of arrowhead; magnitudes shall be indicated.



60.3.1 Application: adjustable



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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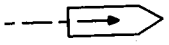
LIST OF SYMBOLS

61 PICKUP HEAD

61.1 ‡ General



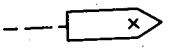
61.2 ‡ Writing; Recording; Head, sound recorder [F]



61.3 ‡ Reading; Playback; Head, sound reproducer [F]



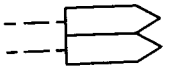
61.4 ‡ Erasing; Eraser, magnetic [F]



61.5 ‡ Application: writing, reading, and erasing



61.6 ‡ Stereo



62 PIEZOELECTRIC CRYSTAL UNIT (including CRYSTAL UNIT, QUARTZ [F])



63 POLARITY SYMBOL

63.1 Positive

\+ IEC

63.2 Negative

— IEC

64 RECEIVER, TELEPHONE EARPHONE [F] HEARING AID RECEIVER

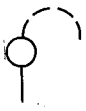
See also HANDSET (item 40).

64.1 General





64.2 Headset, double





64.3 Headset, single





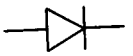
65 RECTIFIER

See ELECTRON TUBE (item 34), SEMICONDUCTOR DIODE (item 73.9.1) and SEMICONDUCTOR DEVICE (item 73).

65.1 General

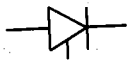
Note 65.1A—Triangle points in direction of forward (easy) current as indicated by a direct current ammeter, unless otherwise noted adjacent to the symbol. Electron flow is in the opposite direction.

Note 65.1B — This symbol represents any method of rectification (electron tube, solid state device, electrochemical device, etc.).



IEC See note 65.1A and B

65.1.1 Controlled



See note 65.1A and B

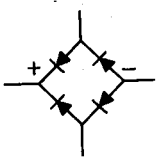
65.2 Semiconductor diode; metallic rectifier; electrolytic rectifier; asymmetrical varistor

The triangle in this case shall be filled.



See note 65.1A

65.2.1 Application: fullwave bridge-type rectifier

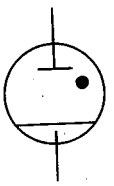


65.3 On connection or wiring diagrams, rectifier may be shown with terminals and polarity marking. Heavy line may be used to indicate nameplate or positive polarity end.



FOR CONNECTION OR WIRING DIAGRAM

65.4 Pool-type-cathode power rectifier



IEC

‡ *The broken line - - - indicates where line connection to a symbol is made and is not a part of the symbol.*

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

66 RELAY

See also CONTACTOR (item 24).

Fundamental symbols for contacts, mechanical connections, coils, etc., are the basis of relay symbols and should be used to represent relays on complete diagrams.

The following letter combinations or symbols may be used with any relay symbols. The requisite number of these letters or symbols may be used to show what special features a relay possesses.

Terms "slow" and "fast" are relative and the degree is not to be noted by multiplicity of same symbol on a relay. Relays that are DC operated are not marked.

	AC	Alternating-current or ringing relay
	D	Differential
	DB	Double biased (biased in both directions)
	DP	Dashpot
	EP	Electrically polarized
	†FO	Fast operate
	†FR	Fast release
	MG	Marginal
	NB	No bias
	NR	Nonreactive
	P	Magnetically polarized using biasing spring, or having magnet bias
	SA	Slow operate and slow release
	SO	Slow operate
	SR	Slow release
	SW	Sandwich wound to improve balance to longitudinal currents

The proper poling for a polarized relay shall be shown by the use of + and — designations applied to the winding leads. The interpretation of this shall be that current in the direction indicated shall move or tend to move the armature toward the contact shown nearest the coil on the diagram. If the relay is equipped with numbered terminals, the proper terminal numbers shall also be shown.

† Used where unusually fast operation or fast releasing is essential to the circuit operation.

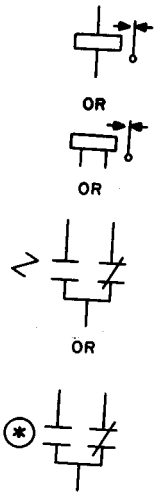
66.1 Basic



66.2 Relay coil

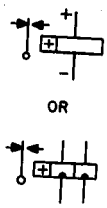
See item 16: COIL, OPERATING; COIL, RELAY.

66.3 Application: relay with transfer contacts

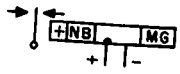


* See note 16A

66.4 Application: polarized relay with transfer contacts



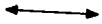
66.5 Application: polarized (no bias) marginal relay with transfer contacts



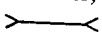
66.6 Relay protective functions

The following symbols may be used to indicate protective functions, or device-function numbers may be placed in the circle or adjacent to the basic symbol (see American Standard C37.2).

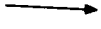
66.6.1 Over, general



66.6.2 Under, general



66.6.3 Direction, general; directional over



66.6.4 Balance, general

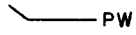


66.6.5 Differential, general

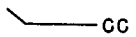
Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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66.6.6 Pilot wire, general



66.6.7 Carrier current, general



66.6.8 Operating quantity

The operating quantity is indicated by the following letters or symbols placed either on or above the center of the relay protective-function symbols shown above.

C	*Current	W	Power
Z	Distance	S	Synchronism
F	Frequency	T	Temperature
GP	Gas pressure	V	Voltage
ϕ	Phase		

* The use of the letter may be omitted in the case of current and the absence of such letter presupposes that the relay operates on current.

66.6.9 Ground relays

Relays operative on residual current only are so designated by attaching the ground symbol  to the relay protective-function symbol. Note that the zero phase-sequence designation given below may be used instead when desirable.

66.6.10 Phase sequence quantities

Operation on phase-sequence quantities may be indicated by the use of the conventional subscripts 0, 1, and 2 after the letter indicating the operating quantity.

66.6.11 Applications

66.6.11.1 Overcurrent



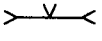
66.6.11.2 Directional overcurrent



66.6.11.3 Directional residual overcurrent



66.6.11.4 Undervoltage



66.6.11.5 Power directional



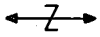
66.6.11.6 Balanced current



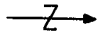
66.6.11.7 Differential current



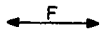
66.6.11.8 Distance



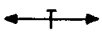
66.6.11.9 Directional distance



66.6.11.10 Overfrequency



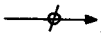
66.6.11.11 Overtemperature



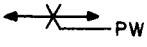
66.6.11.12 Phase balance



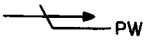
66.6.11.13 Phase sequence



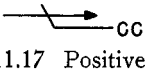
66.6.11.14 Pilot wire, differential current



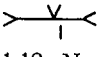
66.6.11.15 Pilot wire, directional comparison



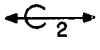
66.6.11.16 Carrier pilot



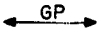
66.6.11.17 Positive phase sequence undervoltage



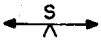
66.6.11.18 Negative phase sequence overcurrent



66.6.11.19 Gas-pressure (Bucholz)



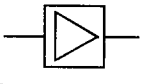
66.6.11.20 Out-of-step

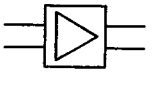


67 REPEATER (includes REPEATER, TELEPHONE[F])

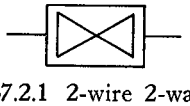
67.1 1-way repeater

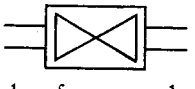
Triangle points in the direction of transmission.





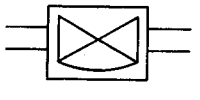
67.2 2-wire 2-way repeater





67.2.1 2-wire 2-way repeater with low-frequency by-pass





Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

67.3 4-wire 2-way repeater



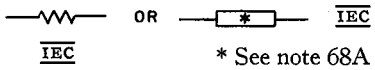
68 RESISTOR

See also TERMINATION (item 82).

For resistors with nonlinear characteristics, see BALLAST LAMP (item 44.1), THERMISTOR (item 84).

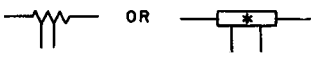
Note 68A—The asterisk is not a part of the symbol. Always add identification within or adjacent to the rectangle.

68.1 General



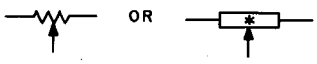
* See note 68A

68.2 Tapped resistor



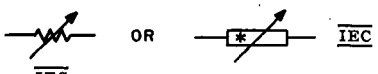
* See note 68A

68.3 Application: with adjustable contact



* See note 68A

68.4 Application: adjustable or continuously adjustable (variable) resistor



* See note 68A

68.5 Heating resistor



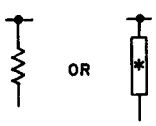
* See note 68A

68.6 Instrument or relay shunt

Connect instrument or relay to terminals in the rectangle.

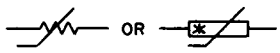


68.7 Shunt resistor



* See note 68A

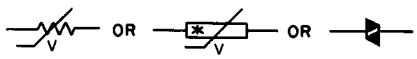
68.8 Nonlinear resistor



* See note 68A

68.8.1 Symmetrical varistor

Resistor, voltage sensitive (silicon carbide, etc.)



* See note 68A

69 RADIATION-SENSITIVITY INDICATOR

69.1 Non-ionizing electromagnetic radiation

This symbol indicates that the principal characteristic of the associated element is designed to vary primarily under the influence of non-ionizing electromagnetic radiation (such as radio waves; infrared, visible or ultraviolet light).



69.2 Ionizing electromagnetic radiation

This symbol indicates that the principal characteristic of the associated element is designed to vary primarily under the influence of ionizing electromagnetic radiation (such as X rays and gamma rays).



69.3 Ionizing particulate radiation

This symbol indicates that the principal characteristic of the associated element is designed to vary primarily under the influence of ionizing particulate radiation. This radiation includes charged particles, such as high-energy electrons (for example, beta rays), alpha rays, and protons; and also neutral particles, such as neutrons and neutral mesons.



70 RESERVED

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

71 RESONATOR CAVITY, TUNED

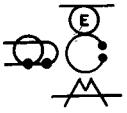
Excluding piezoelectric and magnetostriction devices.

71.1 General
Commonly used for coaxial and waveguide transmission.

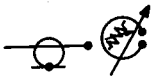


71.2 Applications

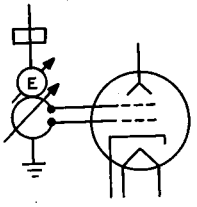
71.2.1 Resonator with mode suppression coupled by an *E*-plane aperture to a guided transmission path and by a loop to a coaxial path



71.2.2 Tunable resonator having adjustable *Q* coupled by a probe to a coaxial system



71.2.3 Tunable resonator with direct-current ground connected to an electron device and adjustably coupled by an *E*-plane aperture to a rectangular waveguide



72 ROTARY JOINT (COUPLER, ROTARY, RADIO FREQUENCY)

72.1 General: with rectangular waveguide system
Note 72.1A—If necessary a transmission path recognition symbol may be added. See item 58. The asterisk is not part of the symbol.



* See note 72.1A

72.2 Coaxial type in rectangular waveguide system



72.3 Circular waveguide type in rectangular waveguide system



73 SEMICONDUCTOR DEVICE TRANSISTOR

See also ENVELOPE; ENCLOSURE (item 35): and TERMINALS (item 81): (for electron tubes, semiconductor devices, etc.).

Note 73A—The equilateral triangle shall be filled and shall touch the semiconductor base region symbol.

Note 73B—Arrowheads on both the N and P emitter symbols shall be half the length of the arrow away from the semiconductor base region.

Note 73C—Emitter, collector and transition line symbols shall be drawn at approximately 60° to the semiconductor base region symbol.

Note 73D—The short lines used in transition symbols shall be appreciably shorter than collector or emitter symbols.

73.1 Semiconductor region with one ohmic connection
As shown, the horizontal line is the semiconductor region and the vertical line is an ohmic connection.

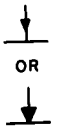


73.1.1 Semiconductor region with a plurality of ohmic connections
(Examples show 2 ohmic connections.)

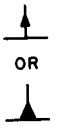


73.2 Rectifying junction or junction which influences a depletion layer

73.2.1 P region on N region



73.2.2 N region on P region



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

73.3 Emitter on region of dissimilar conductivity type
As shown, the slant line with arrow represents the emitter.

73.3.1 P emitter on N region



73.3.1.1 Plurality of P emitters on N region

See note 73B



73.3.2 N emitter on P region



73.3.2.1 Plurality of N emitters on P region

See note 73B



73.4 Collector on region of dissimilar conductivity type
As shown, the slant line represents the collector.



73.4.1 Plurality of collectors on region of dissimilar conductivity type



73.5 Transition between regions of dissimilar conductivity types either P to N, or N to P
The short slant line indicates point of change along the horizontal line from P to N or N to P. No connection shall be made to the short slant line.



73.6 Intrinsic region between 2 regions
The intrinsic region lies *between* the linked slant lines.

73.6.1 Between regions of dissimilar conductivity type, either PIN or NIP



73.6.2 Between regions of similar conductivity type, either PIP or NIN



73.6.3 Between a collector and a region of dissimilar conductivity type, either PIN or NIP
The connection to the collector is made to the long slant line.



73.6.4 Between a collector and a region of similar conductivity type, either PIP or NIN
The connection to the collector is made to the long slant line.



73.7 Special properties
Note 73.7A—If necessary, a special function or property, essential for circuit operation may be indicated by a supplementary symbol placed within envelope, or adjacent to the symbol.

73.7.1 Radiation (electromagnetic or particulate) dependence
See item 69.

73.7.2 t° Temperature dependence

73.7.3  Capacitive device

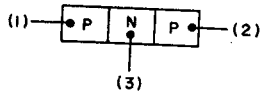
73.7.4  Tunneling device

73.7.5  Breakdown device

Note 73.7.5A—Do not show in mirror-image form.

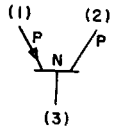
73.8 Rules for drawing semiconductor device symbols
Note 73.8A—To draw a device symbol, start at an electrode whose polarity is known (usually an emitter) and proceed along the device, showing all of its regions individually. Finally, indicate ohmic connections where required.

73.8.1 PNP Transistor (example of a 3 element device)



ACTUAL DEVICE

Construction of symbol by successively using items 73.3.1, 73.4, and 73.1.

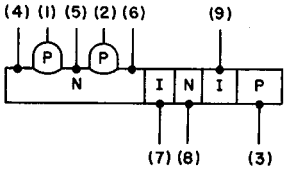


SYMBOL
(LETTERS AND NUMBERS
ARE NOT A PART OF THE SYMBOL)

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

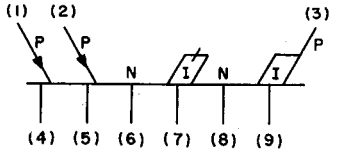
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

73.8.2 PNINIP (example of a multi-element device)



ACTUAL DEVICE

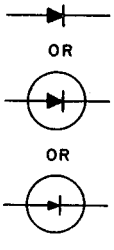
Construction of symbol by successively using items 73.3.1.1, 73.6.2, 73.6.3, and 73.1.1.



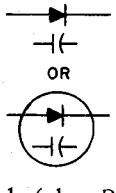
SYMBOL
(LETTERS AND NUMBERS ARE NOT A PART OF THE SYMBOL)

73.9 Typical applications; two terminal devices (See notes in item 73.10.)

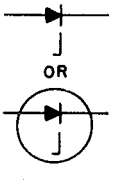
73.9.1 Semiconductor diode
Semiconductor rectifier diode



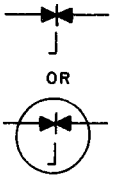
73.9.2 Capacitive diode (Varactor)



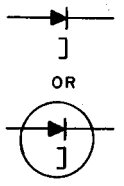
73.9.3 Breakdown diode, unidirectional (also Backward diode)



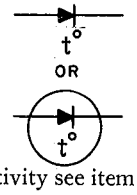
73.9.4 Breakdown diode, bidirectional



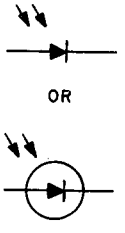
73.9.5 Tunnel diode



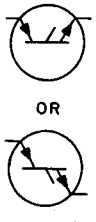
73.9.6 Temperature dependent diode



73.9.7 Photodiode (for radiation sensitivity see item 69)



73.9.8 Semiconductor diode, PNPN-type switch



73.10 Typical applications; three or more terminal devices

73.10A In general, the angle at which a lead is brought to a symbol element has no significance.

73.10B The envelope symbol may be omitted if no confusion would arise or if none of the elements is connected to the envelope.

73.10C Orientation, including a mirror-image presentation, does not change the meaning of a symbol.

73.10D The elements of the symbol must be drawn in such an order as to show clearly the operating function of the device.

73.10.1 PNP transistor (also PNIP transistor, if omitting the intrinsic region will not result in ambiguity)



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

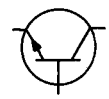
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LIST OF SYMBOLS

73.10.1.1 Application: PNP transistor with one electrode connected to envelope (in this case the collector electrode)
See note 73.10B



73.10.2 NPN transistor (also NPIN transistor, if omitting the intrinsic region will not result in ambiguity)



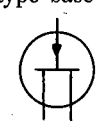
73.10.3 Unijunction transistor with N-type base



73.10.4 Unijunction transistor with P-type base



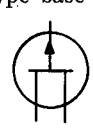
73.10.5 Field-effect transistor with N-type base



OR



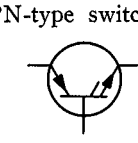
73.10.6 Field-effect transistor with P-type base



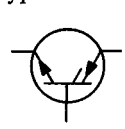
OR



73.10.7 Semiconductor triode, PNPN-type switch



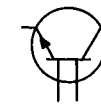
73.10.8 Semiconductor triode, NPNP-type switch



73.10.9 NPN transistor with transverse-biased base



OR



73.10.10 PNIP transistor with ohmic connection to the intrinsic region



73.10.11 NPIN transistor with ohmic connection to the intrinsic region



73.10.12 PNIN transistor with ohmic connection to the intrinsic region



73.10.13 NPIP transistor with ohmic connection to the intrinsic region



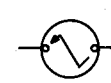
74 SHIELD
SHIELDING (*short dashes*)

Normally used for electric or magnetic shielding. When used for other shielding, a note should so indicate. For typical applications see:
CAPACITOR (item 8.1.2).
PATH, TRANSMISSION (items 58.8.1, 58.8.4, and 58.8.6).
TRANSFORMER (items 86.2.1 and 86.2.2).

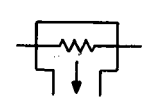
— — — — —

75 SQUIB

75.1 Explosive



75.2 Igniter



75.3 Sensing link; Fusible link, ambient temperature operated



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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Check the source to verify that this is the current version before use.

76 SWITCH [E]

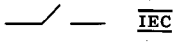
See also FUSE (item 36); CONTACT, ELECTRIC (item 23); and DRAFTING PRACTICES (paragraphs 3.7 and 3.9).

Fundamental symbols for contacts, mechanical connections, etc., may be used for switch symbols.

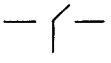
The standard method of showing switches is in a position with no operating force applied. For switches that may be in any one of two or more positions with no operating force applied and for switches actuated by some mechanical device (as in air-pressure, liquid-level, rate-of-flow, etc., switches), a clarifying note may be necessary to explain the point at which the switch functions.

When the basic switch symbols in items 76.1 through 76.4 are shown on a diagram in the closed position, terminals must be added for clarity.

76.1 Single throw, general



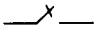
76.2 Double throw, general



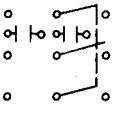
76.2.1 Application: 2-pole double-throw switch with terminals shown



76.3 Knife switch, general

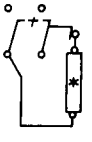


76.3.1 Application: 3-pole double-throw knife switch with auxiliary contacts and terminals



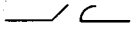
76.3.2 Application: 2-pole field-discharge knife switch with terminals and discharge resistor

Note 76.3.2A—The asterisk is not a part of the symbol. Always add identification within or adjacent to the rectangle.

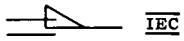


* See note 76.3.2A

76.4 Switch with horn gap



76.5 Sector switch



76.6 Push button, momentary or spring return

76.6.1 Circuit closing (make)



76.6.2 Circuit opening (break)



76.6.3 Two-circuit



76.7 Push button, maintained or not spring return

76.7.1 Two circuit

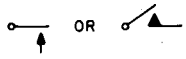


76.8 Switch, nonlocking; momentary or spring return

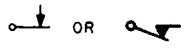
The symbols to the left are commonly used for spring buildups in key switches, relays, and jacks.

The symbols to the right are commonly used for toggle switches.

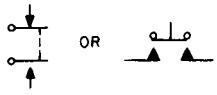
76.8.1 Circuit closing (make)



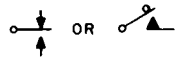
76.8.2 Circuit opening (break)



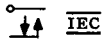
76.8.3 Two-circuit



76.8.4 Transfer



76.8.5 Make-before-break



76.9 Switch, locking

The symbols to the left are commonly used for spring buildups in key switches, relays, and jacks.

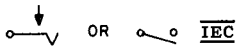
The symbols to the right are commonly used for toggle switches.

Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

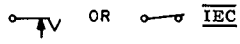
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

LIST OF SYMBOLS

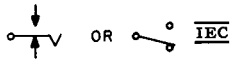
76.9.1 Circuit closing (make)



76.9.2 Circuit opening (break)



76.9.3 Transfer, 2-position



76.9.4 Transfer, 3-position



76.9.5 Make-before-break

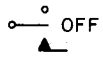


76.10 Switch, combination locking and nonlocking

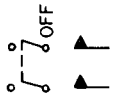
See also item 76.11.

Commonly used for toggle-switches.

76.10.1 3-position 1-pole: circuit closing (make), off, momentary circuit closing (make)



76.10.2 3-position 2-pole: circuit closing (make), off, momentary circuit closing (make)

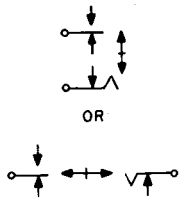


76.11 Switch, key-type, applications

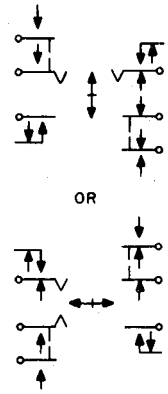
76.11.1 2-position with locking transfer and break contacts



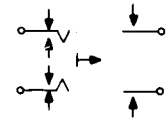
76.11.2 3-position with nonlocking transfer and locking break contacts



76.11.3 3-position, multicontact combination



76.11.4 2-position, half of key switch normally operated, multicontact combination

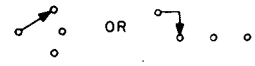


76.12 Selector or multiposition switch

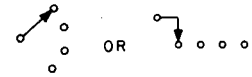
The position in which the switch is shown may be indicated by a note or designation of switch position.

76.12.1 General (for power and control diagrams)

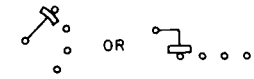
Any number of transmission paths may be shown.



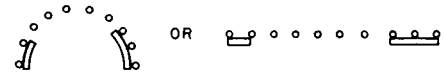
76.12.2 Break-before-make, nonshorting (nonbridging) during contact transfer



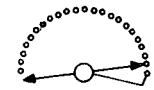
76.12.3 Make-before-break, shorting (bridging) during contact transfer



76.12.4 Segmental contact



76.12.5 22-point selector switch



76.12.6 10-point selector switch with fixed segment



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

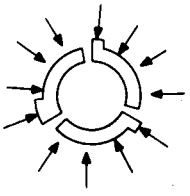
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Check the source to verify that this is the current version before use.

76.12.7 Wafer, 3-pole 3-circuit with 2 nonshorting and 1 shorting moving contacts

Viewed from end opposite control knob or actuator unless otherwise indicated.

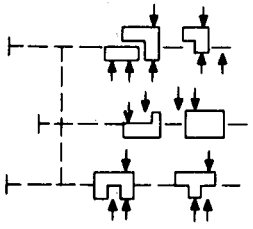
For more than one section, section No. 1 is nearest control knob.

When contacts are on both sides, front contacts are nearest control knob.



76.12.8 Slide switch, typical ladder-type interlock

In the example, one slide is shown operated. Slides are shown in released position unless otherwise noted.



76.12.9 Master or control switch

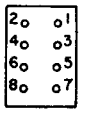
A table of contact operation must be shown on the diagram. A typical table is shown below.

DETACHED CONTACTS SHOWN ELSEWHERE ON DIAGRAM

CONTACT	POSITION		
	A	B	C
1-2			X
3-4	X		
5-6			X
7-8	X		

X INDICATES CONTACT CLOSED

HANDLE END



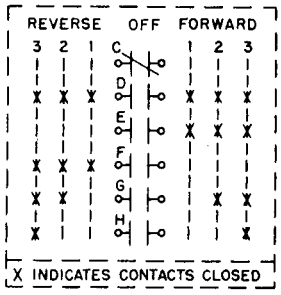
FOR CONNECTION OR WIRING DIAGRAM

76.12.10 Master or control switch

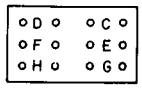
(Cam-operated contact assembly) 6-circuit 3-point reversing switch.

A table of contact operation must be shown on the diagram. A typical table is shown below. Tabulate special features in note.

DETACHED CONTACTS SHOWN ELSEWHERE ON DIAGRAM

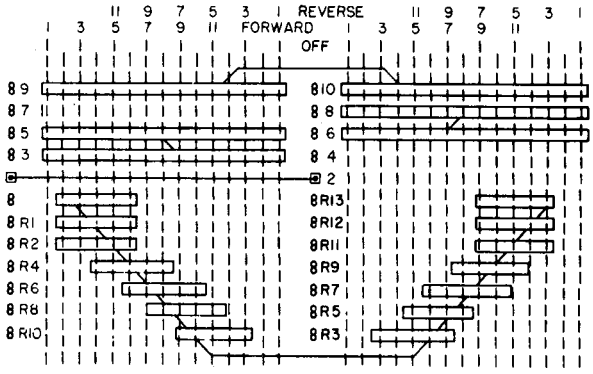


HANDLE END



FOR CONNECTION OR WIRING DIAGRAM

76.12.11 Drum switch, sliding-contact type, typical example



76.13 Switches with specific features

76.13.1 Key-operated lock switch

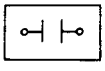
Use appropriate standard symbol and add key designation or other information in note.

76.13.2 Limit switch

Note 76.13.2A—Identify by LS or other suitable note.

76.13.2.1 Track-type, circuit-closing contact

See note 76.13.2A

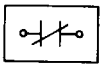


Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

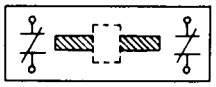
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LIST OF SYMBOLS

76.13.2.2 Track-type, circuit-opening contact

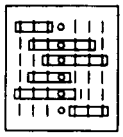
See note 76.13.2A 

76.13.2.3 Lead-screw type, circuit-opening contacts



See note 76.13.2A

76.13.2.4 Rotary type



See note 76.13.2A

76.13.3 Mushroom-head safety feature
Application to 2-circuit push-button switch.



76.13.4 Safety interlock

76.13.4.1 General
If specific type identification is not required, use applicable standard symbol.

76.13.4.2 If specific type identification is required; circuit opening



76.13.4.3 If specific type identification is required; circuit closing

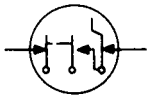


76.13.5 Switch, hook F



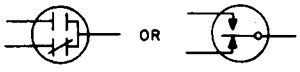
76.13.6 Telephone dial F, (switch)





TYPICAL

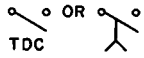
76.13.7 Switch in evacuated envelope, 1-pole double-throw



76.13.8 Switches with time delay feature (see also items 23.4 and 23.5)

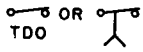
Note 76.13.8A—The point of the arrow indicates the direction of switch operation in which contact action is delayed.

76.13.8.1 Open switch with time delay closing (TDC) feature



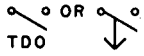
See note 76.13.8A

76.13.8.2 Closed switch with time delay opening (TDO) feature



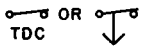
See note 76.13.8A

76.13.8.3 Open switch with time delay opening (TDO) feature



See note 76.13.8A

76.13.8.4 Closed switch with time delay closing (TDC) feature



See note 76.13.8A

76.13.9 Flow actuated switch

76.13.9.1 Closing on increase in flow



76.13.9.2 Opening on increase in flow



76.13.10 Liquid level actuated switch

76.13.10.1 Closing on rising level



76.13.10.2 Opening on rising level



76.13.11 Pressure or Vacuum actuated switch

76.13.11.1 Closing on rising pressure

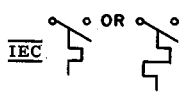


76.13.11.2 Opening on rising pressure



76.13.12 Temperature actuated switch

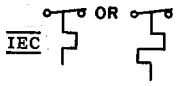
76.13.12.1 Closing on rising temperature



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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76.13.12.2 Opening on rising temperature



76.13.13 Foot operated switch

76.13.13.1 Opened by foot pressure



76.13.13.2 Closed by foot pressure



76.13.14 Switch operated by shaft rotation and responsive to speed or direction (F = forward, R = reverse) see also item 37

76.13.14.1 Speed



76.13.14.2 Plugging: to stop drive after it has come practically to rest



76.13.14.3 Anti-plugging: to prevent plugging of drive



76.13.15 Limit switch, direct actuated, spring returned

76.13.15.1 Normally open



76.13.15.2 Normally open-held closed



76.13.15.3 Normally closed



76.13.15.4 Normally closed-held open



77 SWITCHING FUNCTION

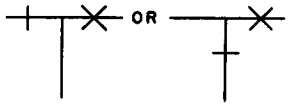
77.1 Conducting, closed contact (break)



77.2 Nonconducting, open contact (make)



77.3 Application: transfer

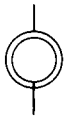


78 SYNCHRO F

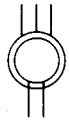
If identification is required, a letter combination from the following list shall be placed adjacent to the symbol to indicate the type of synchro.

CDX	Control-differential transmitter
CT	Control transformer
CX	Control transmitter
TDR	Torque-differential receiver
TDX	Torque-differential transmitter
TR	Torque receiver
TX	Torque transmitter
RS	Resolver

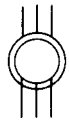
If the outer winding is rotatable in bearings, the suffix B shall be added to the above letter combinations.



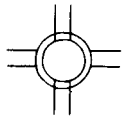
78.1 Synchro, control transformer F
Synchro, receiver F
Synchro, transmitter F



78.2 Synchro, differential receiver F
Synchro, differential transmitter F



78.3 Synchro, resolver F
Type shown, 2 phase rotor and 2 phase stator.



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

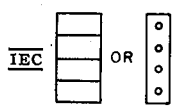
79 RESERVED

80 TERMINAL, CIRCUIT F
See also TUBE TERMINALS (item 81.2).

○ IEC

80.1 Terminal board F or terminal strip, with 4 terminals shown; group of 4 terminals

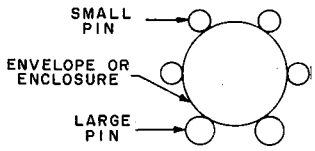
Number and arrangement as convenient.



81 TERMINALS (for electron tubes, semiconductor devices, etc.)
Primarily used in application-data terminal diagrams for electron tubes, semiconductor devices, and other devices having terminations of similar type. See also CONNECTOR (item 18).

Note 81A—Explanatory words and arrows are not a part of the symbol.

81.1 Base pin terminals (electron tubes, etc.); pin terminals (semiconductor devices, etc.)



See note 81A

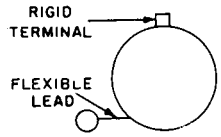
81.1.1 The following letter combinations, if shown adjacent to terminal symbols requiring special attention, shall signify the following:

S—Connection to an external shield integral with a device (including metal tube shell, base sleeve or shell; external conductive coating or casing). Not to be used if the external conductive coating serves as one side of a capacitor (as in cathode-ray tubes) and is not designed to function as an electrostatic shield.

IC—Terminal not intended to be used for circuit connection.

IS—Internal shield not depicted in terminal diagram.

81.2 Envelope terminals



See note 81A

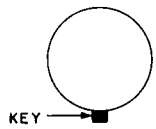
The rigid-terminal symbol is used to indicate customary rigid terminals (caps, rods, rings, etc.) as well as to indicate:

(a) Any metallic envelope or external conductive coating or casing that has a contact area (as in cathode-ray tubes, disc-seal tubes, pencil tubes, etc.)

(b) Mounting flange or stud when it serves as a terminal.

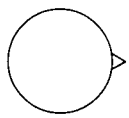
81.3 Terminal group orientation

81.3.1 Device with base orientation key

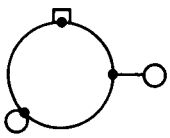


See note 81A

81.3.2 Devices with reference point (such as a boss, colored dot, index pin, index tab, or bayonet pin).



81.4 Terminals connected to metallic envelope or enclosure



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

81.5 Applications

Note 81.5A—If the device terminals are in a circular arrangement, the actual angular spacing between the terminals should be approximated on the terminal diagram.

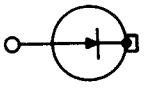
Note 81.5B—If the terminals are in an essentially linear arrangement the terminal diagram may show the terminals in either a linear array along one side of the elongated envelope symbol (preferable), or within a maximum angle of 150° around the circular envelope symbol.

Note 81.5C—If pins are omitted in an otherwise standard terminal arrangement, do not respace the remaining pins.

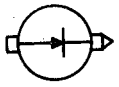
Note 81.5D—A terminal at the center of the terminal arrangement shall be identified as the CENTER terminal lead or pin.

Note 81.5E—The typical examples show pin numbering in accordance with standard industry practice, i.e., with the terminals viewed from outside the terminal face of the device.

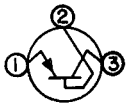
81.5.1 Two-terminal device with one flexible lead and one rigid terminal connected to a metallic envelope (typical semiconductor diode shown).



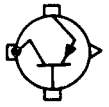
81.5.2 Two-terminal device with rigid terminals and reference point located at one of the terminals (typical semiconductor diode shown).



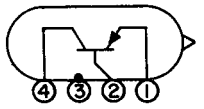
81.5.3 Three-terminal device with circular arrangement of pin terminals with base orientation determined by gap in pin spacing (typical transistor shown).



81.5.4 Three-terminal device with rigid terminals, one connected to the metallic enclosure, and index pin (typical transistor shown).



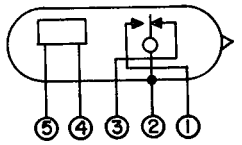
81.5.5 Four-terminal device with in-line pin terminals, one connected to metallic envelope, and reference point (typical transistor shown).



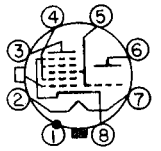
OR



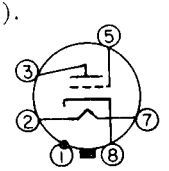
81.5.6 Five-terminal device with in-line terminal leads, one connected to metallic enclosure and reference point (typical relay shown).



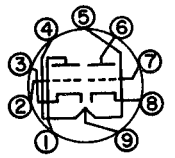
81.5.7 Device with 8-terminal keyed (such as octal) base, rigid envelope terminal, and metallic envelope connected to base terminal (typical triode-heptode shown).



81.5.8 Device with keyed (such as octal) base having design capability of 8 pins but with 2 pins omitted, and with 3 rigid envelope terminals (typical disc-seal triode shown).



81.5.9 Device with 9-terminal (such as noval) base utilizing gap in pin spacing to establish base orientation (typical twin triode shown).



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

82 TERMINATION

82.1 Cable termination

Line on left of symbol shown indicates cable.



82.2 Open circuit (open)

Not a fault. Commonly used in coaxial and waveguide diagrams.

82.3 Short circuit (short)

Not a fault. Commonly used in coaxial and waveguide diagrams.



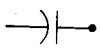
82.3.1 Application: movable short



82.4 Terminating capacitor

Commonly used in coaxial and waveguide diagrams.

82.4.1 Application: series capacitor and path open



82.4.2 Application: series capacitor and path shorted



82.5 Terminating inductor

Commonly used in coaxial and waveguide diagrams.

82.5.1 Application: series inductor and path open



82.5.2 Application: series inductor and path shorted



82.6 Terminating resistor

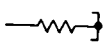
Commonly used in coaxial and waveguide diagrams.



82.6.1 Application: series resistor and path open



82.6.2 Application: series resistor and path shorted



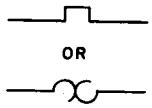
83 THERMAL ELEMENT

(Thermomechanical transducer)

83.1 Actuating device

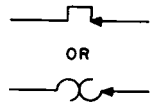
May be either self or externally heated.

Note 83.1A—Use appropriate number of single-line diagram symbols.



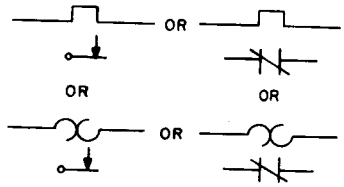
See note 83.1A

83.2 Thermal cutout; flasher



See note 83.1A

83.3 Thermal relay



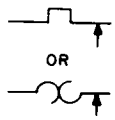
83.4 Thermostat

Ambient-temperature-operated device.

Operates on rising temperature.

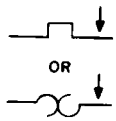
83.4.1 With break contact

See also item 76.13.12.2

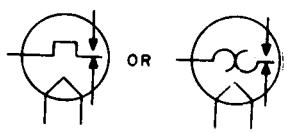


83.4.2 With make contact

See also item 76.13.12.1



83.4.3 With integral heater and transfer contacts



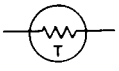
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LIST OF SYMBOLS

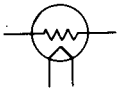
84 THERMISTOR
RESISTOR, THERMAL F

"T" indicates that the primary characteristic of the element within the circle is a function of temperature.

84.1 General



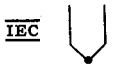
84.2 With independent integral heater



85 THERMOCOUPLE F

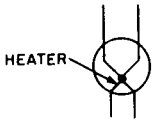
85.1 Dissimilar-metals device

85.1.1 Temperature-measuring thermocouple

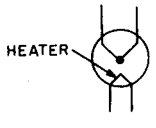


85.1.2 Current-measuring thermocouple
Explanatory words and arrows are not a part of the symbols shown.

85.1.2.1 Thermocouple with integral heater internally connected

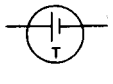


85.1.2.2 Thermocouple with integral insulated heater

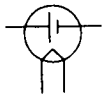


85.2 Semiconductor device

85.2.1 Temperature-measuring semiconductor thermocouple



85.2.2 Current-measuring semiconductor thermocouple



86 TRANSFORMER F

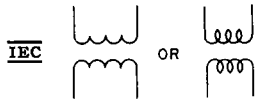
86.1 General

Either winding symbol may be used. In the following items, the left symbol is used.

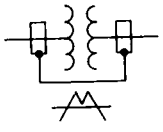
Additional windings may be shown or indicated by a note.

For polarity markings on current and potential transformers, see items 86.16.1 and 86.17.1.

In coaxial and waveguide circuits, this symbol will represent a taper or step transformer without mode change.



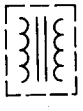
86.1.1 Application: transformer with direct-current connections and mode suppression between two rectangular waveguides



86.2 If it is desired especially to distinguish a magnetic-core transformer



86.2.1 Application: shielded transformer with magnetic core shown



86.2.2 Application: transformer with magnetic core shown and with a shield between windings. The shield is shown connected to the frame



86.3 One winding with adjustable inductance



86.4 Each winding with separately adjustable inductance



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

86.5 Adjustable mutual inductor, constant-current transformer



86.6 With taps, 1-phase



86.7 Autotransformer, 1-phase



86.7.1 Adjustable



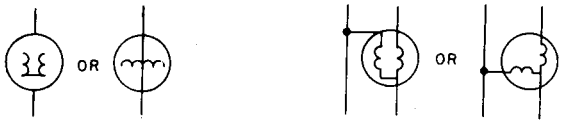
86.8 Step-voltage regulator or load-ratio control autotransformer



86.9 Load-ratio control transformer with taps



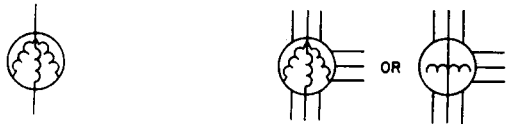
86.10 1-phase induction voltage regulator(s)
Number of regulators may be written adjacent to the symbol.



86.11 Triplex induction voltage regulator



86.12 3-phase induction voltage regulator



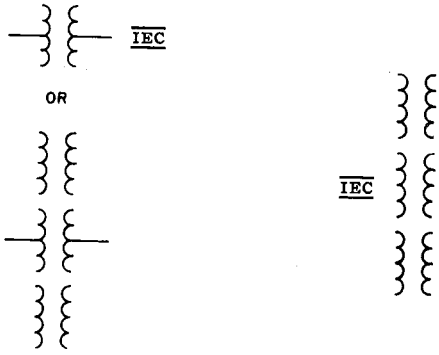
86.13 1-phase 2-winding transformer



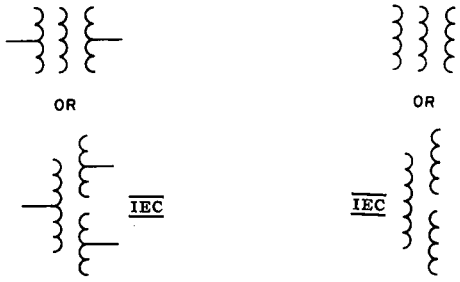
86.13.1 3-phase bank of 1-phase 2-winding transformers



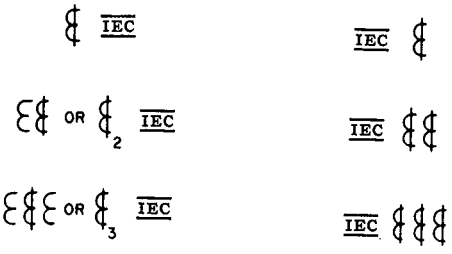
86.14 Polyphase transformer



86.15 1-phase 3-winding transformer



86.16 Current transformer(s)

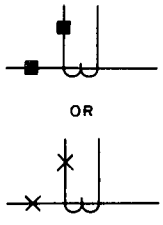


Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

86.16.1 Current transformer with polarity marking
Instantaneous direction of current into one polarity mark corresponds to current out of the other polarity mark.

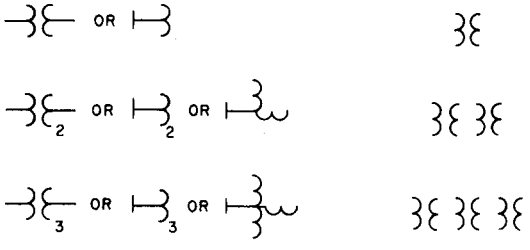


Symbol used shall not conflict with item 77 when used on same drawing.

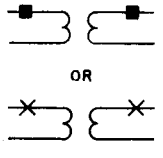
86.16.2 ‡ Bushing-type current transformer



86.17 Potential transformer(s)

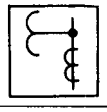


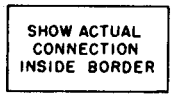
86.17.1 Potential transformer with polarity mark
Instantaneous direction of current into one polarity mark corresponds to current out of the other polarity mark.



Symbol used shall not conflict with item 77 when used on same drawing.

86.18 Outdoor metering device





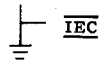
‡*The broken line - —— indicates where line connection to a symbol is made and is not a part of the symbol.*

86.19 Transformer winding connection symbols
For use adjacent to the symbols for the transformer windings.

86.19.1 2-phase 3-wire, ungrounded



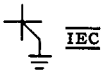
86.19.1.1 2-phase 3-wire, grounded



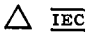
86.19.2 2-phase 4-wire



86.19.2.1 2-phase 5-wire, grounded



86.19.3 3-phase 3-wire, delta or mesh



86.19.3.1 3-phase 3-wire, delta, grounded



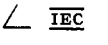
86.19.4 3-phase 4-wire, delta, ungrounded



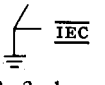
86.19.4.1 3-phase 4-wire, delta, grounded



86.19.5 3-phase, open-delta



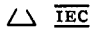
86.19.5.1 3-phase, open-delta, grounded at common point



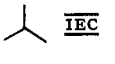
86.19.5.2 3-phase, open-delta, grounded at middle point of one transformer



86.19.6 3-phase, broken-delta



86.19.7 3-phase, wye or star, ungrounded



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

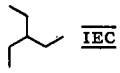
86.19.7.1 3-phase, wye, grounded neutral
The direction of the stroke representing the neutral can be arbitrarily chosen.



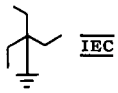
86.19.8 3-phase 4-wire, ungrounded



86.19.9 3-phase, zigzag, ungrounded



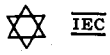
86.19.9.1 3-phase, zigzag, grounded



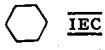
86.19.10 3-phase, Scott or T



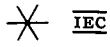
86.19.11 6-phase, double-delta



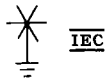
86.19.12 6-phase, hexagonal (or chordal)



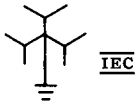
86.19.13 6-phase, star (or diametrical)



86.19.13.1 6-phase, star, with grounded neutral

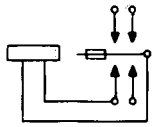


86.19.14 6-phase, double zigzag with neutral brought out and grounded

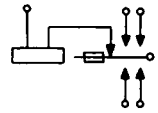


87 VIBRATOR, INTERRUPTER F

87.1 Typical shunt drive (with terminals shown)
Show contacts as required.



87.2 Typical separate drive (with terminals shown)
Show contacts as required.



88 VISUAL SIGNALING DEVICE

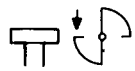
88.1 Annunciator F (general)



88.1.1 Annunciator drop or signal, shutter or grid type



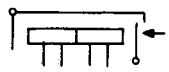
88.1.2 Annunciator drop or signal, ball type



88.1.3 Manually restored drop



88.1.4 Electrically restored drop



88.2 Communication switchboard-type lamp



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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LIST OF SYMBOLS

88.3 Indicating, pilot, signaling, or switchboard light
Light, indicator
Light, signal

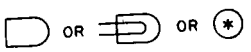
See also LAMP (item 44).

If confusion with other circular symbols may occur, the D-shaped symbol should be used.

Note 88.3A—To indicate the following characteristics, the specified letter or letters may be inserted within or placed adjacent to the symbol.

A	Amber
B	Blue
C	Clear
G	Green
NE	Neon
O	Orange
OP	Opalescent
P	Purple
R	Red
W	White
Y	Yellow

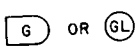
Note 88.3B—The asterisk is not part of the circular symbol. Always add the letter or letters specified in Note 88.3A within or adjacent to the circle. To avoid confusion with meter or basic relay symbols, add suffix L or IL to the above letter or letters; for example, RL or RIL placed within or adjacent to the circle.



See note 88.3A

* See note 88.3B

88.3.1 Application: green signal light



88.4 Jeweled signal light



See note 88.3A

89 RESERVED

90 RESERVED

91 RESERVED

92 RESERVED

93 RESERVED

94 RESERVED

95 RESERVED

96 RESERVED

97 RESERVED

98 RESERVED

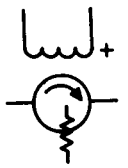
99 RESERVED

100 FERRITE DEVICES

100.1 Field polarization rotator
Arrow indicates direction of rotation of electric field when viewed in direction of signal flow.



100.2 Field polarization amplitude modulator



Note—Single-line (one-line) symbols appear at the left, complete symbols at the right, and symbols suitable for both purposes are centered in each column.

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Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Copies of this standard for military use may be obtained as indicated in the general provisions of the Department of Defense Index of Specifications and Standards.

The title and identifying symbol should be stipulated when requesting copies of military standards.

Custodians:
Army—Sig C
Navy—Ships
Air Force—AFLC

Preparing activity:
Navy—Ships
Project DRPR-0063

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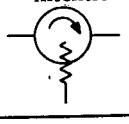
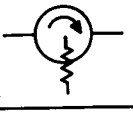
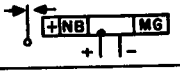
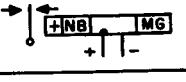
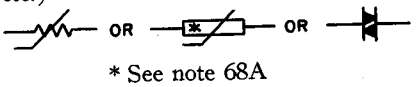
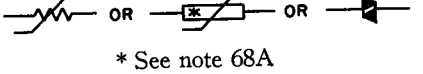
APPENDIX A

REVISED OR DELETED SYMBOLS

(RELATING TO MIL-STD-15-1, DATED 30 OCTOBER 1961)

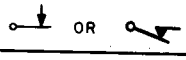
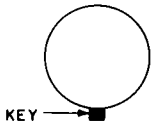
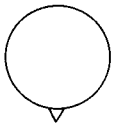
10. REVISED OR DELETED SYMBOLS

10.1 This appendix is added for information only.

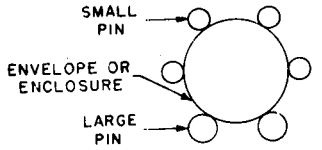
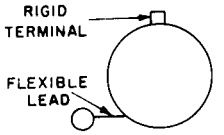
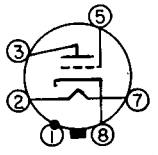
Revised or Deleted Symbols in MIL-STD-15-1 (including original item numbers).	Recommended Symbols in Standard MIL-STD-15-1A if not otherwise specified.
EXPANDED 1 ADJUSTABLE CONTINUOUSLY ADJUSTABLE (Variable) The shaft of the arrow is drawn at about 45 degrees across the body of the symbol.  IEC	See ADJUSTABILITY, VARIABILITY (item 1)
REVISED 9 CELL, PHOTOSENSITIVE (Semiconductor) See also PHOTOTUBE (item 34.11.6).  indicates that the primary characteristic of the element within the circle is designed to vary under the influence of light.	See RADIATION - SENSITIVITY INDICATOR (item 69)
CORRECTION 57.4 Unidirectional (Isolator) Power flowing in direction of arrow is not intentionally attenuated. 	57.4 Unidirectional (Isolator) Power flowing in direction of arrow is not intentionally attenuated. 
CORRECTION 66.5 Application: polarized (no bias) marginal relay with transfer contacts 	66.5 Application: polarized (no bias) marginal relay with transfer contacts 
CORRECTION 68.8.1 Symmetrical varistor Resistor, voltage sensitive F (silicon carbide, etc.)  * See note 68A	68.8.1 Symmetrical varistor Resistor, voltage sensitive F (silicon carbide, etc.)  * See note 68A

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APPENDIX A—Continued

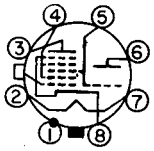
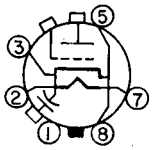
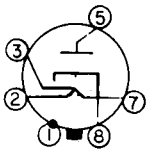
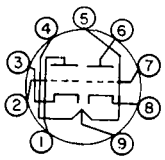
Revised or Deleted Symbols in MIL-STD-15-1 (including original item numbers).	Recommended Symbols in Standard MIL-STD-15-1A if not otherwise specified.
REVISED 73.7.1  Light dependence	See RADIATION - SENSITIVITY INDICATOR (item 69)
CORRECTION 76.8.2 Circuit opening (break) 	76.8.2 Circuit opening (break) 
CORRECTION 76.13.14.2 Plugging: to stop drive after it has practically come to rest 	76.13.14.2 Plugging: to stop drive after it has come practically to rest 
EXPANDED 81 TERMINAL AND BASING CONNECTIONS FOR CONNECTION (WIRING) DIAGRAMS Not normally used for schematic diagrams. 81.1 Basing orientation symbols 81.1.1 For tubes with keyed bases Explanatory word and arrow are not a part of the symbol shown.  81.1.2 For tubes with bayonets, bosses, and other reference points 	See TERMINALS (for electron tubes, semiconductor devices, etc.) (item 81)

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Revised or Deleted Symbols in MIL-STD-15-1 (including original item numbers).	Recommended Symbols in Standard MIL-STD-15-1A if not otherwise specified.
EXPANDED (Continued) 81.2 Tube terminals The usage of the rigid-envelope-terminal symbol of item 81.2.2 includes the indication of any external metallic envelope or conducting coating or casing that has a contact area (as ir cathode-ray tubes, metallic "pencil" tubes, etc.). However, where contact to such external metallic elements is made through a base terminal, a dot junction is employed as in item 81.3.1 to indicate that voltage applied to this base terminal may make the envelope alive. Terminal symbols may be added to the composite device symbols where desired without changing the meaning or becoming a part of the symbol. 81.2.1 Base terminals Explanatory words and arrows are not a part of the symbol.  81.2.2 Envelope terminals Explanatory words and arrows are not a part of the symbol.  81.3 Applications 81.3.1 Triode with indirectly heated cathode and envelope connected to base terminal  	See TERMINALS (for electron tubes, semiconductor devices, etc.) (item 81)

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APPENDIX A—Continued

Revised or Deleted Symbols in MIL-STD-15-1 (including original item numbers).	Recommended Symbols in Standard MIL-STD-15-1A if not otherwise specified.
EXPANDED (Continued) 81.3.2 Triode-heptode with rigid envelope connection  81.3.3 Ultra-high-frequency triode (disk-seal-tube type) with internal capacitor  81.3.4 Rectifier with heater tap and envelope connected to base terminal  81.3.5 Equipotential-cathode twin triode with tapped heater 	See TERMINALS (for electron tubes, semiconductor devices, etc.) (item 81)

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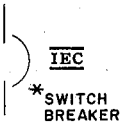
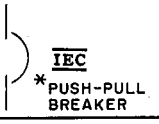
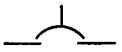
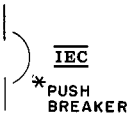
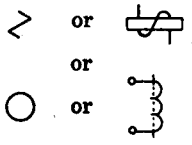
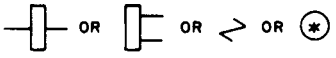
APPENDIX B

REVISED OR DELETED SYMBOLS

(RELATING TO MIL-STD-15A, DATED 1 APRIL 1954)

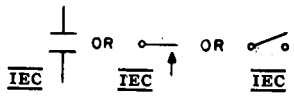
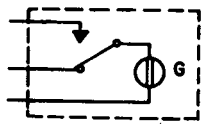
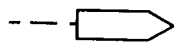
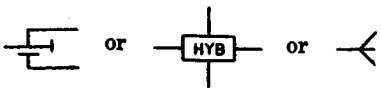
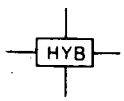
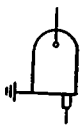
20. REVISED OR DELETED SYMBOLS

20.1 This appendix is added for information only.

Revised or Deleted Symbols in MIL-STD-15A (including original item numbers).	Recommended Symbols in Standard MIL-STD-15-1A if not otherwise specified.
DELETED **3.9.2** ***Switch breaker.*** 	See CIRCUIT BREAKER (item 11)  * Not a part of the symbol
DELETED **3.9.3** ***Push-pull breaker.*** 	See CIRCUIT BREAKER (item 11)  * Not a part of the symbol
DELETED **3.9.4** ***Push breaker.*** 	See CIRCUIT BREAKER (item 11)  * Not a part of the symbol
REVISED **3.11 COIL, OPERATING (BASIC).**  **In all cases there shall be a designation within the circle.**	See COIL, OPERATING (item 16) For buildups using coil symbol see also items 6.4, 16.1, 16.2, 26.1, 26.2, 66.3, 66.4, 66.5, 87.1, 87.2, 88.1.1, 88.1.2, 88.1.3, and 88.1.4.  * See note 16A.
DELETED (BUILDUP) **3.13.9.3** ***Three-conductor (jack).*** 	See CONTACT, ELECTRICAL (item 23) 

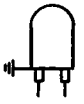
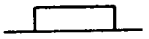
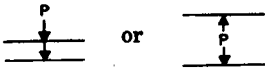
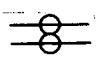
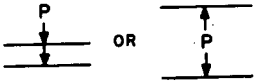
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APPENDIX B—Continued

Revised or Deleted Symbols in MIL-STD-15A (including original item numbers).	Recommended Symbols in Standard MIL-STD-15-1A if not otherwise specified.
DELETED **3.14.12** ***Buildup example:*** **Latch contact, electrically-operated, mechanically-held device.** **For aircraft drawings or where specified.** 	See Open contact (make) (item 23.3.2) and include explanation 
DELETED **3.21.10** ***Buildup example:*** **Green jeweled indicating light with push-to-test circuit.** 	See Switch (item 76.8.1) and Indicator light (item 88.3) 
REVISED **3.30 HEAD, REPRODUCER, SOUND; PICKUP (MECHANO-ELECTRIC).**  Suitable words or abbreviations may be written within or adjacent to the rectangle.	See PICKUP (item 61) 
REVISED AND DELETED **3.31.1** ***Hybrid, general.*** 	Deleted two alternates, see Hybrid (item 41.1) 
DELETED (BUILDUP) **3.35.7.2** ***Synchronous motor or generator or condenser, 3-phase wye (ungrounded).*** 	See 3-phase wye (ungrounded) (item 46.7.3) and Synchronous motor (item 46.10.1) 
DELETED **3.35.8.1** ***One low tension terminal, one booster terminal, frame grounded.*** 	See Magnetoelectric generator, 1 phase (item 46.9.10) 

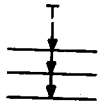
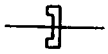
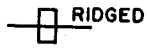
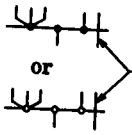
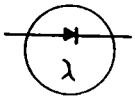
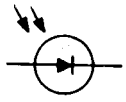
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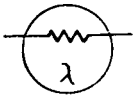
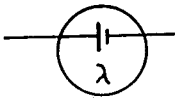
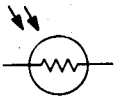
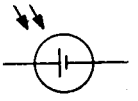
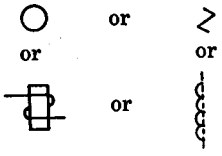
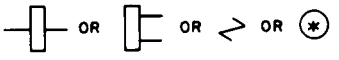
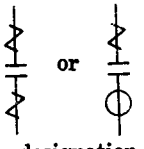
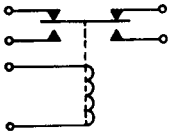
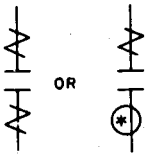
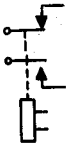
Revised or Deleted Symbols in MIL-STD-15A (including original item numbers).	Recommended Symbols in Standard MIL-STD-15-1A if not otherwise specified.
DELETED 3.35.8.2 *Two low tension terminals and frame grounded.* 	See item 46.9.10 
DELETED 3.35.9 *Hand generator.* 	See item 46.9.10 
DELETED **3.42.7.6 *Wire and cable identification:* if identification is required.** L48C20N **or** L48C20N **The arrow may be omitted if no confusion will result.**	Wiring diagram requirement not a part of Standard MIL-STD-15-1A.
DELETED **3.42.7.6.1 *Cable marker for cable termination.***  **For wiring diagrams (when required by the procuring agency).** **Identification of receptacle to which the plug on the end of the cable connects may be placed within or adjacent to the rectangle.**	Wiring diagram requirement not a part of Standard MIL-STD-15-1A.
REVISED **3.42.7.9 *Pair.***  **3.42.7.10 *Twisted pair.*** 	See Path, transmission (items 58.2, 58.7.1 and 58.7.2) 

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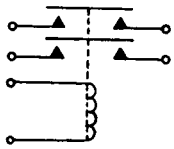
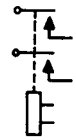
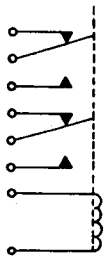
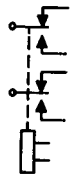
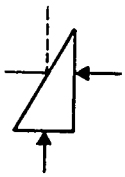
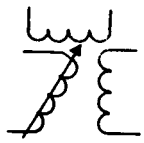
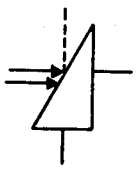
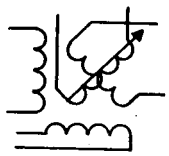
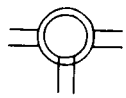
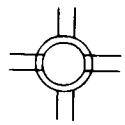
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REVISED (Continued) **3.42.7.11** *Triple.*  **3.42.7.12** *Twisted triple.* 	
DELETED **3.42.12.3** *Ridged.* 	See Waveguide (item 58.11) 
DELETED **3.42.13** *Bus.*  Crossing conductor; no connection The line for bus may be emphasized.	See Drafting Practices (paragraphs 3.5 and 3.11)
REVISED **3.43 PHOTOELECTRIC CELL.** For Phototube, see par. 3.64.11.6. "λ" indicates that the primary characteristics of the element within the circle is designed to vary under the influence of light. **3.43.1** *Asymmetrical photoconductive transducer (resistive).* The arrowhead in this case shall be filled. 	See Radiation-Sensitivity indicator (item 69)  Non-ionizing electromagnetic radiation See Cell, photosensitive (items 9.1, 9.2 and 9.3) 

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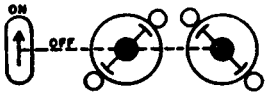
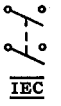
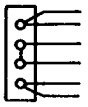
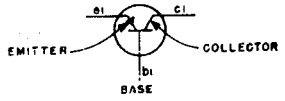
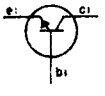
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REVISED (Continued) **3.43.2** *Selenium cell; Symmetrical photoconductive transducer.*  **3.43.3** ***Barrier photocell; Blocking layer cell; Photovoltaic transducer.*** 	 
REVISED **3.47.2** *Relay coil.*  In all cases there shall be a designation within the circle.	See Coil, operating (item 16) For buildups using coil symbol see also items 6.4, 16.1, 16.2, 26.1, 26.2, 66.3, 66.4, 66.5, 87.1, 87.2, 88.1.1, 88.1.2, 88.1.3, and 88.1.4.  * See note 16A
DELETED (BUILDUP) **3.47.4.6** *Electrically operated single-pole relay or contactor with shunt blowout coil.*  In all cases there shall be a designation within the circle. 	See item 24.4  * See note 24.2A See COIL, OPERATING (item 16) and Basic contact assemblies (item 23.3) 

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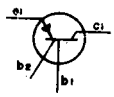
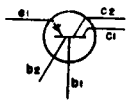
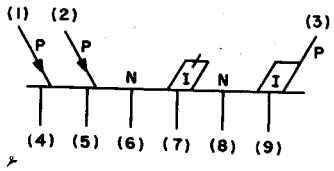
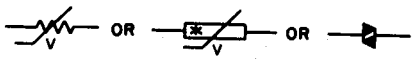
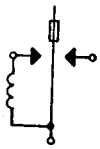
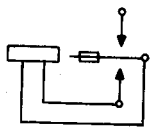
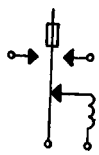
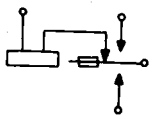
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DELETED (BUILDUP) 3.47.4.7 *Two circuit.* 3.47.4.8 *Double pole, single throw, double make.* 	See COIL, OPERATING (item 16) and Basic contact assemblies (item 23.3) 
DELETED (BUILDUP) 3.47.4.9 *Double pole, double throw.* 	See COIL, OPERATING (item 16) and Basic contact assemblies (item 23.3) 
DELETED **3.50 RESOLVER (ELECTROMECHANICAL).** Arrows denote input in single-line item. 3.50.1 *Singly-wound rotor.*   3.50.2 *Doubly-wound rotor.*  	See Synchro, resolver (item 78.3)  

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DELETED **3.56.15.8** *Rotary snap switch.* 	See Switch, locking (second form) (item 76.9.1) 
DELETED **3.58.2** ***Buildup example:*** **Terminal board with leads and a bus connecting two of the terminals. Bends in leads as shown on the symbol shall not occur at the outline of the terminal board.** 	Drafting practice, see ASA STANDARD Y14.15 
THIS SECTION HAS BEEN EXPANDED **3.63 TRANSISTOR; Electro-electrical transducer.** Explanatory words and leader lines are not part of the symbols. The arrowheads on the emitter shall be filled and not touch line. Letters are not part of symbol but may be used for clarity. **3.63.1** ***Triode;*** **N or PNP type.**  Arrowhead on emitter shows direction of current flow. Explanatory words and leader lines are not part of symbols. **3.63.2** ***Triode;*** **P, NPN or NPNP type.** 	See SEMICONDUCTOR DEVICE (item 73) PNP transistor (also PNIP transistor, if omitting the intrinsic region will not result in ambiguity)  NPN transistor (also NPIN transistor, if omitting the intrinsic region will not result in ambiguity) 

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EXPANDED (Continued) **3.63.3** *Tetrode.*  **3.63.4** *Pentode.* 	Construction of symbol: See item 73.8  SYMBOL (LETTERS AND NUMBERS ARE NOT A PART OF THE SYMBOL)
REVISED **3.65 VARISTOR; RESISTOR, VOLTAGE SENSITIVE.** Also see RECTIFIER par. 3.46. Electro-electrical transducer with non-linear characteristics. The arrowheads in these cases shall be filled. **RELOCATED** **3.65.1** *Asymmetrical:* Metallic rectifier. Arrow shows direction of forward (easy) current as indicated by DC ammeter.  **RELOCATED** **3.65.2** *Symmetrical.* 	See Asymmetrical varistor (item 65.2) Semiconductor diode; metallic rectifier; electrolytic rectifier; asymmetrical varistor The triangle in this case shall be filled.  See note 65.1A See Symmetrical varistor (item 68.8.1) Symmetrical varistor Resistor, voltage sensitive (silicon carbide, etc.) 
REVISED **3.66.1** *Typical, shunt drive (contacts as required).* 	See VIBRATOR, INTERRUPTER (item 87) 
REVISED **3.66.2** *Typical, separate drive (contacts as required).* 	See VIBRATOR, INTERRUPTER (item 87) 

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Note—This index contains commonly accepted or standard item names as used in the list of symbols, as well as some colloquial names. Certain of these names may not appear under the item listing.

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